US012407961B2

(12) **United States Patent**
Niwa et al.

(10) **Patent No.: US 12,407,961 B2**
(45) **Date of Patent: *Sep. 2, 2025**

(54) **SOLID-STATE IMAGING ELEMENT, IMAGING DEVICE, AND CONTROL METHOD OF SOLID-STATE IMAGING ELEMENT**

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(72) Inventors: **Atsumi Niwa**, Kanagawa (JP); **Yusuke Oike**, Kanagawa (JP)

(73) Assignee: **Sony Semiconductor Solutions Corporation**, Kanagawa (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/742,377**

(22) Filed: **Jun. 13, 2024**

(65) **Prior Publication Data**

US 2024/0340552 A1 Oct. 10, 2024

Related U.S. Application Data

(63) Continuation of application No. 18/142,949, filed on May 3, 2023, now Pat. No. 12,177,592, which is a (Continued)

(30) **Foreign Application Priority Data**

Jan. 23, 2018 (JP) 2018-008850

(51) **Int. Cl.**

H04N 25/772	(2023.01)
H04N 25/443	(2023.01)
H04N 25/445	(2023.01)
H04N 25/47	(2023.01)
H04N 25/707	(2023.01)
H04N 25/78	(2023.01)

(52) **U.S. Cl.**

CPC .... ***H04N 25/772*** (2023.01); ***H04N 25/443*** (2023.01); ***H04N 25/445*** (2023.01); (Continued)

(58) **Field of Classification Search**

CPC .. H04N 25/772; H04N 25/443; H04N 25/445; H04N 25/47; H04N 25/53; (Continued)

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(Continued)

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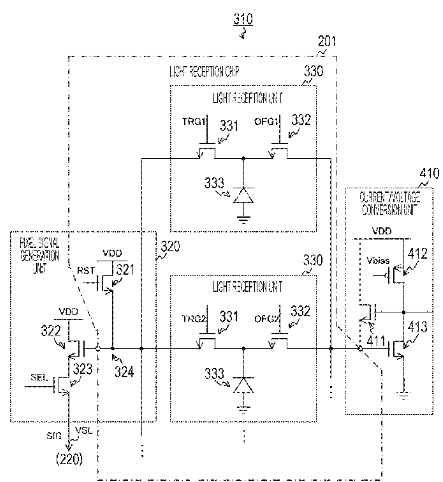
(74) *Attorney, Agent, or Firm* — Michael Best & Friedrich LLP

(57) **ABSTRACT**

An object is to reduce a circuit scale in a solid-state imaging element that detects an address event. The solid-state imaging element is provided with a plurality of photoelectric conversion elements, a signal supply unit, and a detection unit. In this solid-state imaging element, each of the plurality of photoelectric conversion elements photoelectrically converts incident light to generate a first electric signal. Furthermore, in the solid-state imaging element, the detection unit detects whether or not a change amount of the first electric signal of each of the plurality of photoelectric conversion elements exceeds a predetermined threshold and

(Continued)



outputs a detection signal indicating a result of the detection result.

20 Claims, 32 Drawing Sheets

Related U.S. Application Data

continuation of application No. 17/935,343, filed on Sep. 26, 2022, now Pat. No. 11,659,304, which is a continuation of application No. 16/962,783, filed as application No. PCT/JP2019/001517 on Jan. 18, 2019, now Pat. No. 11,470,273.

(52) **U.S. Cl.**
CPC ***H04N 25/47*** (2023.01); ***H04N 25/707*** (2023.01); ***H04N 25/78*** (2023.01)

(58) **Field of Classification Search**
CPC H04N 25/707; H04N 25/75; H04N 25/778; H04N 25/78; H04N 25/79; H04N 25/766; H10F 39/803; H10F 39/12; H10F 39/18; H10F 39/809; H10F 39/813
See application file for complete search history.

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KR	20140103116	A	8/2014
KR	20160071386	A	6/2016
TW	201241995	A	10/2012
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FIG. 1

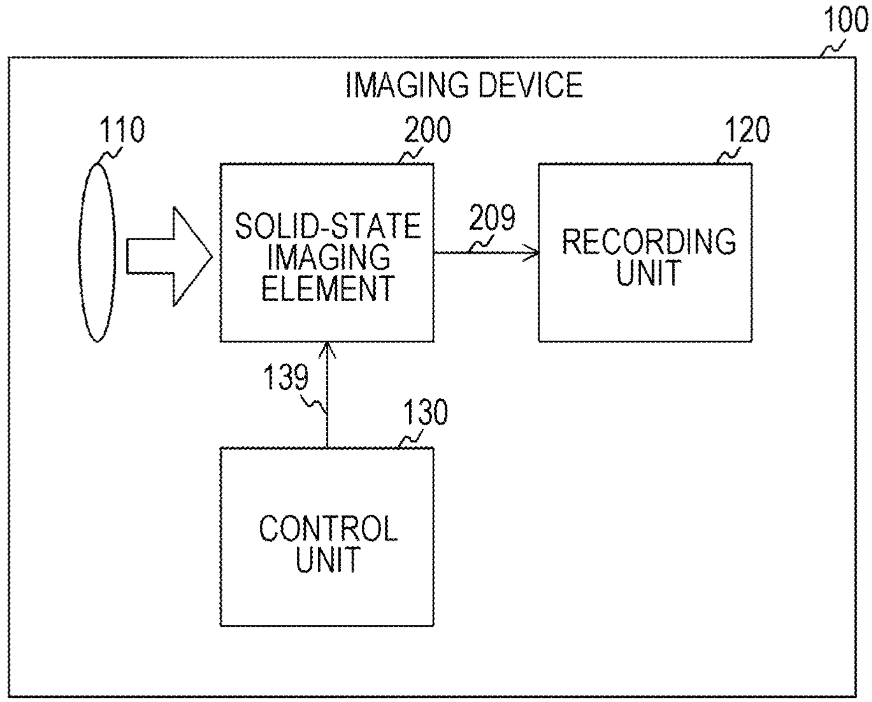

100
IMAGING DEVICE
110
200
SOLID-STATE IMAGING ELEMENT
209
120
RECORDING UNIT
139
130
CONTROL UNIT

FIG. 2

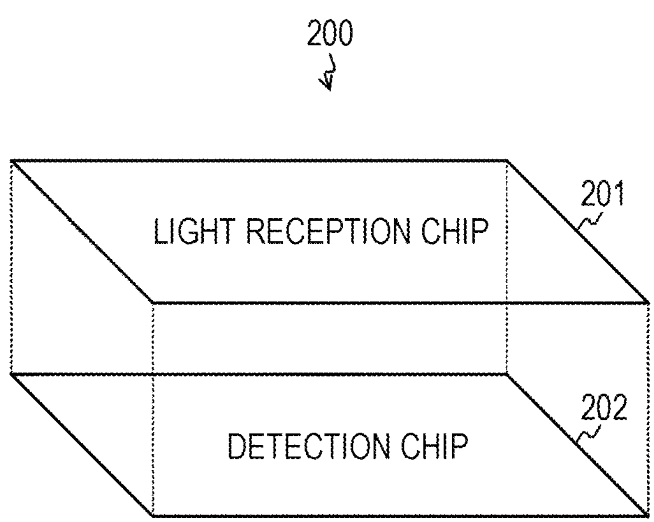
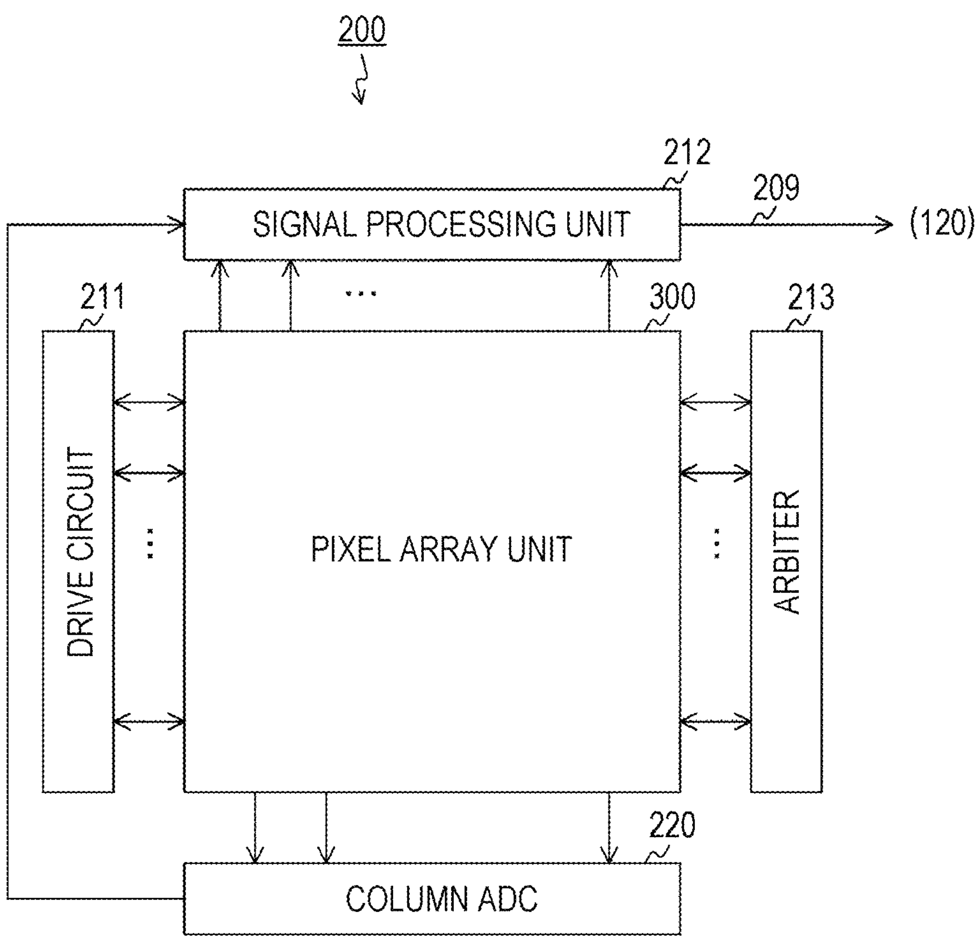

200
201
LIGHT RECEPTION CHIP
202
DETECTION CHIP

FIG. 3



200
212
SIGNAL PROCESSING UNIT
209
(120)
211
300
213
DRIVE CIRCUIT
PIXEL ARRAY UNIT
ARBITER
220
COLUMN ADC

FIG. 4

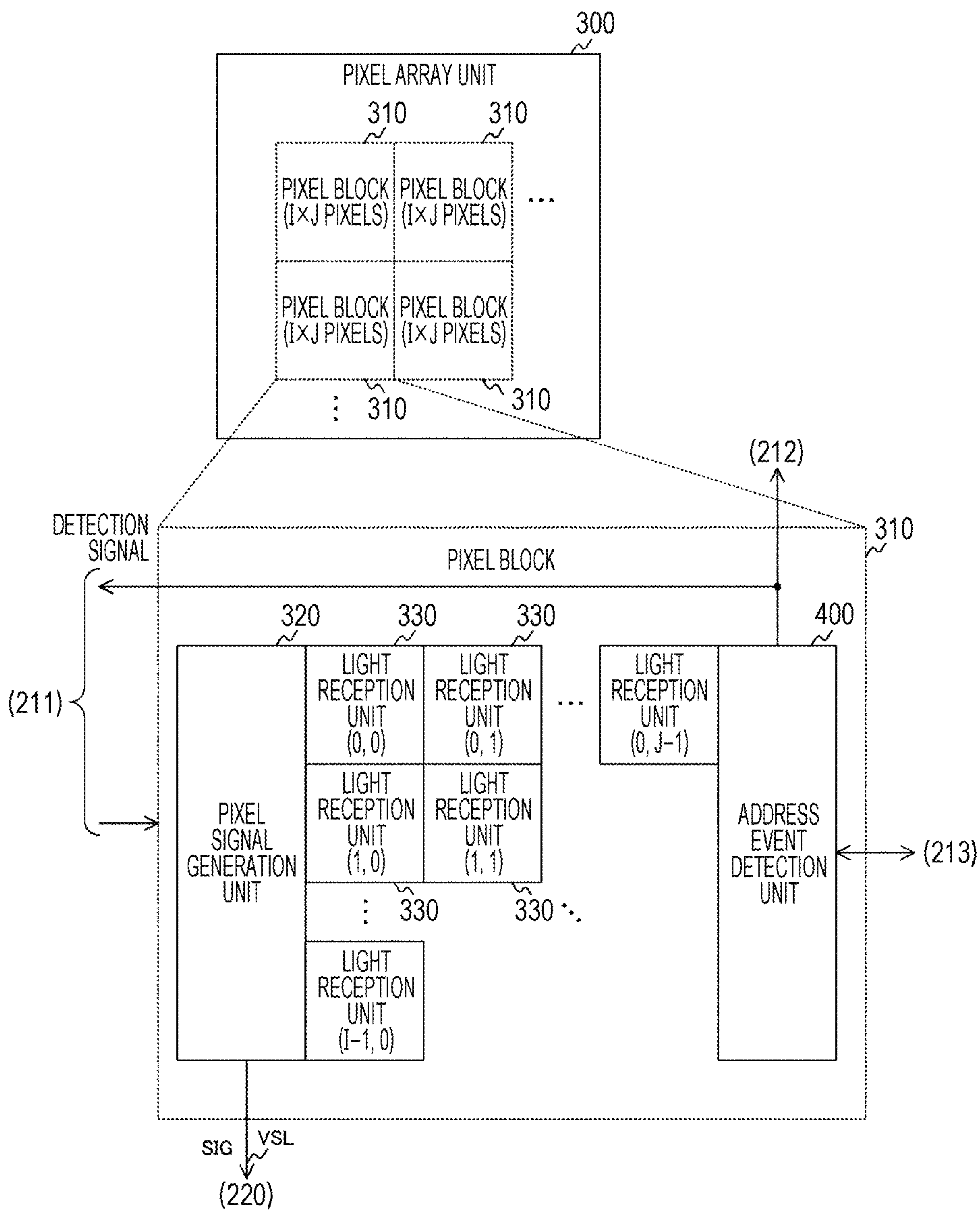
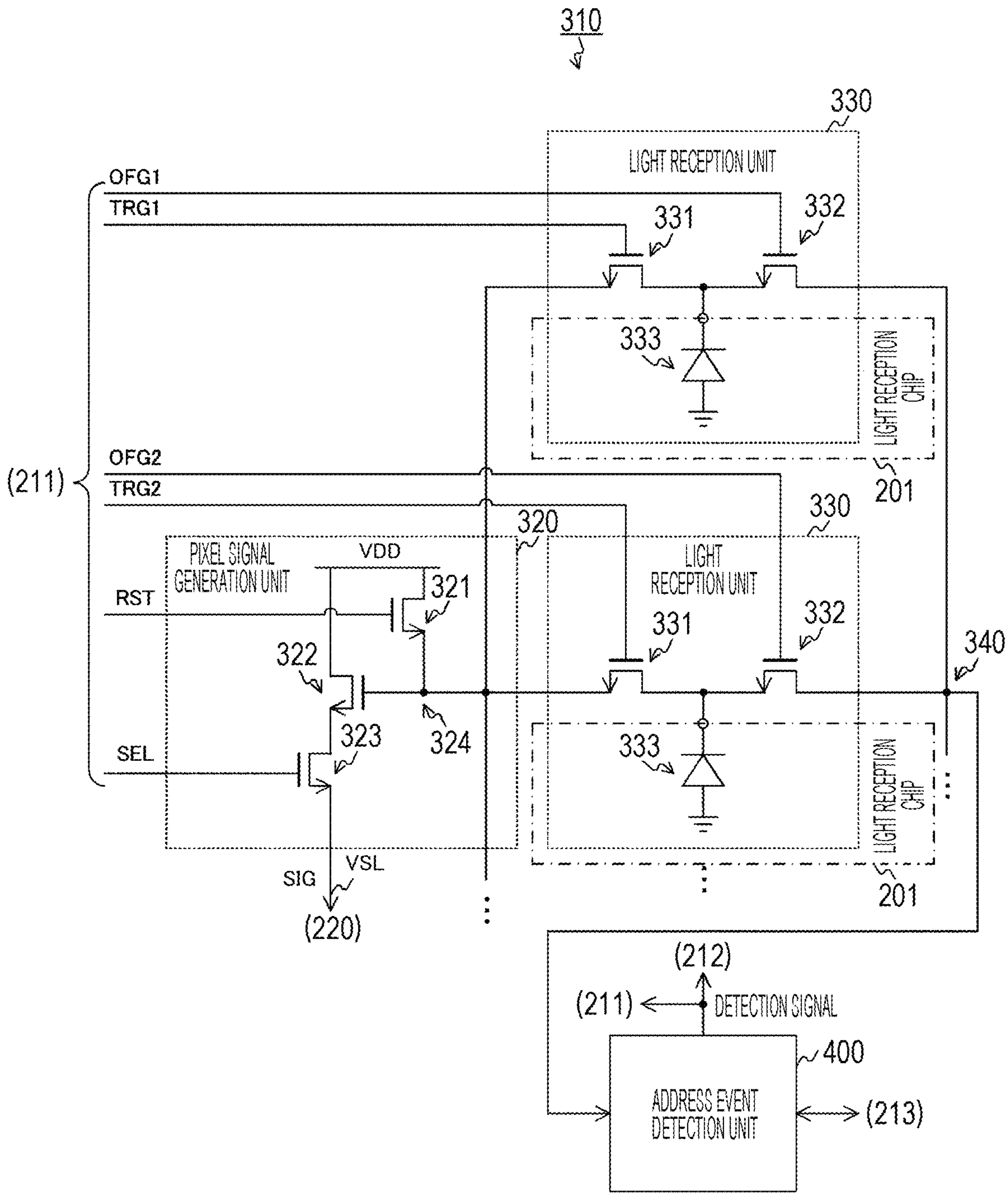

300
PIXEL ARRAY UNIT
310
310
PIXEL BLOCK (I×J PIXELS)
PIXEL BLOCK (I×J PIXELS)
PIXEL BLOCK (I×J PIXELS)
PIXEL BLOCK (I×J PIXELS)
310
310
(212)
DETECTION SIGNAL
310
PIXEL BLOCK
320
330
330
400
(211)
LIGHT RECEPTION UNIT (0, 0)
LIGHT RECEPTION UNIT (0, 1)
LIGHT RECEPTION UNIT (0, J−1)
LIGHT RECEPTION UNIT (1, 0)
LIGHT RECEPTION UNIT (1, 1)
PIXEL SIGNAL GENERATION UNIT
ADDRESS EVENT DETECTION UNIT
(213)
330
330
LIGHT RECEPTION UNIT (I−1, 0)
SIG
VSL
(220)

FIG. 5



310
330
LIGHT RECEPTION UNIT
OFG1
TRG1
331
332
333
LIGHT RECEPTION CHIP
201
(211)
OFG2
TRG2
330
320
PIXEL SIGNAL GENERATION UNIT
VDD
LIGHT RECEPTION UNIT
RST
321
331
332
340
322
323
324
333
SEL
LIGHT RECEPTION CHIP
201
SIG
VSL
(220)
(212)
(211)
DETECTION SIGNAL
400
ADDRESS EVENT DETECTION UNIT
(213)

FIG. 6

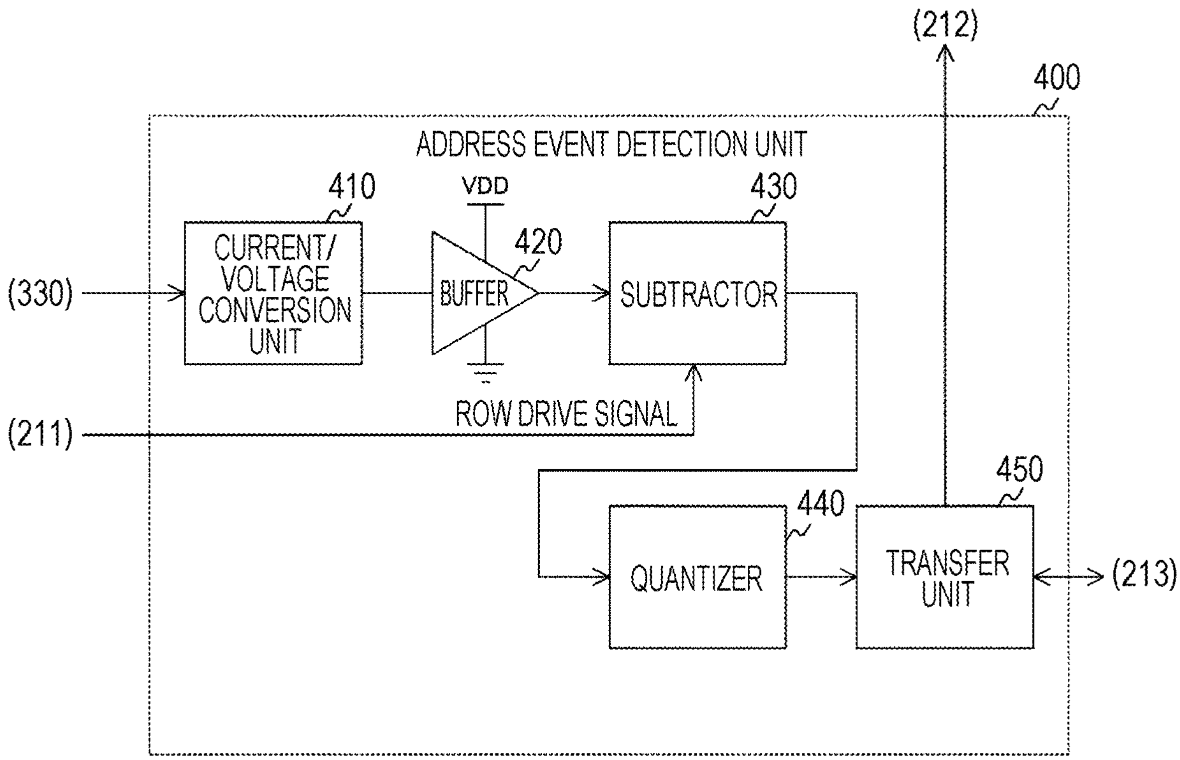

(212)
400
ADDRESS EVENT DETECTION UNIT
410
VDD
430
CURRENT/ VOLTAGE CONVERSION UNIT
420
(330)
BUFFER
SUBTRACTOR
ROW DRIVE SIGNAL
(211)
450
440
QUANTIZER
TRANSFER UNIT
(213)

FIG. 7

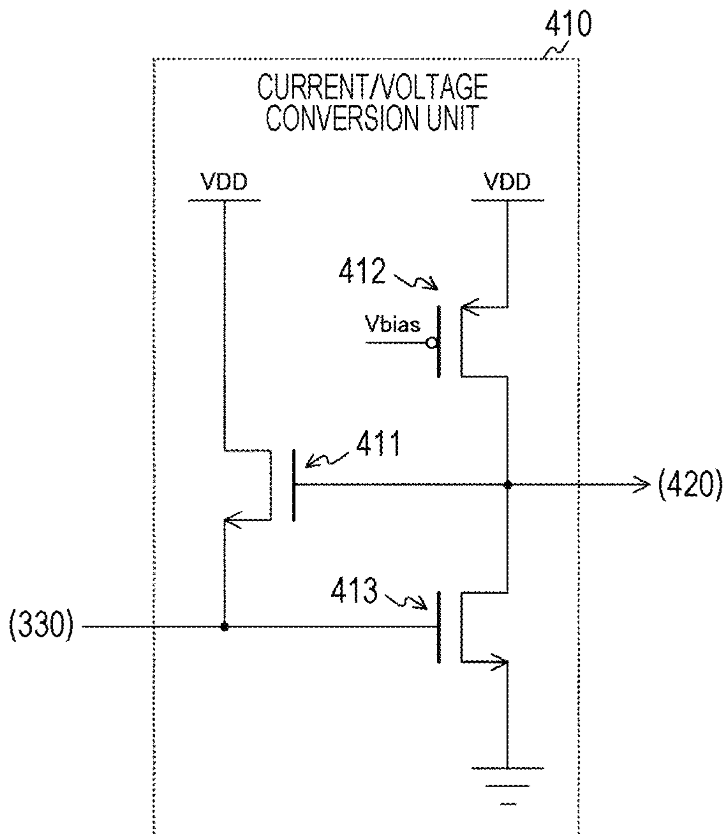

410
CURRENT/VOLTAGE
CONVERSION UNIT
VDD
VDD
412
Vbias
411
(420)
413
(330)

FIG. 8

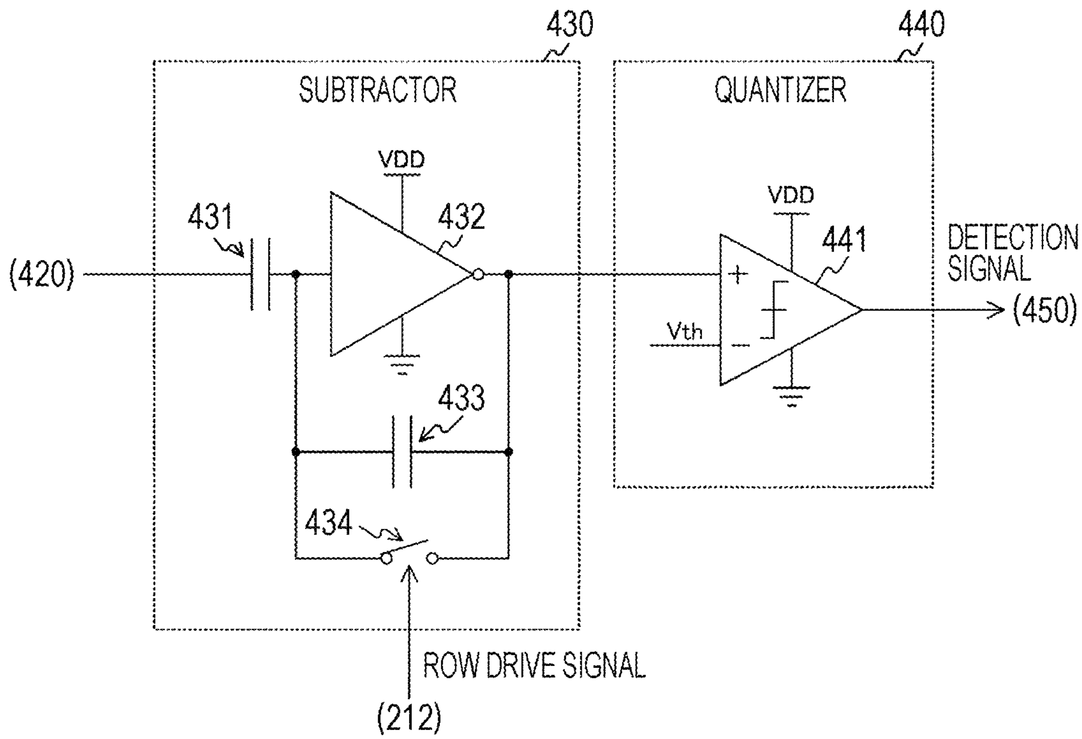

430
440
SUBTRACTOR
QUANTIZER
VDD
VDD
431
432
441
(420)
DETECTION
SIGNAL
(450)
Vth
433
434
ROW DRIVE SIGNAL
(212)

FIG. 9

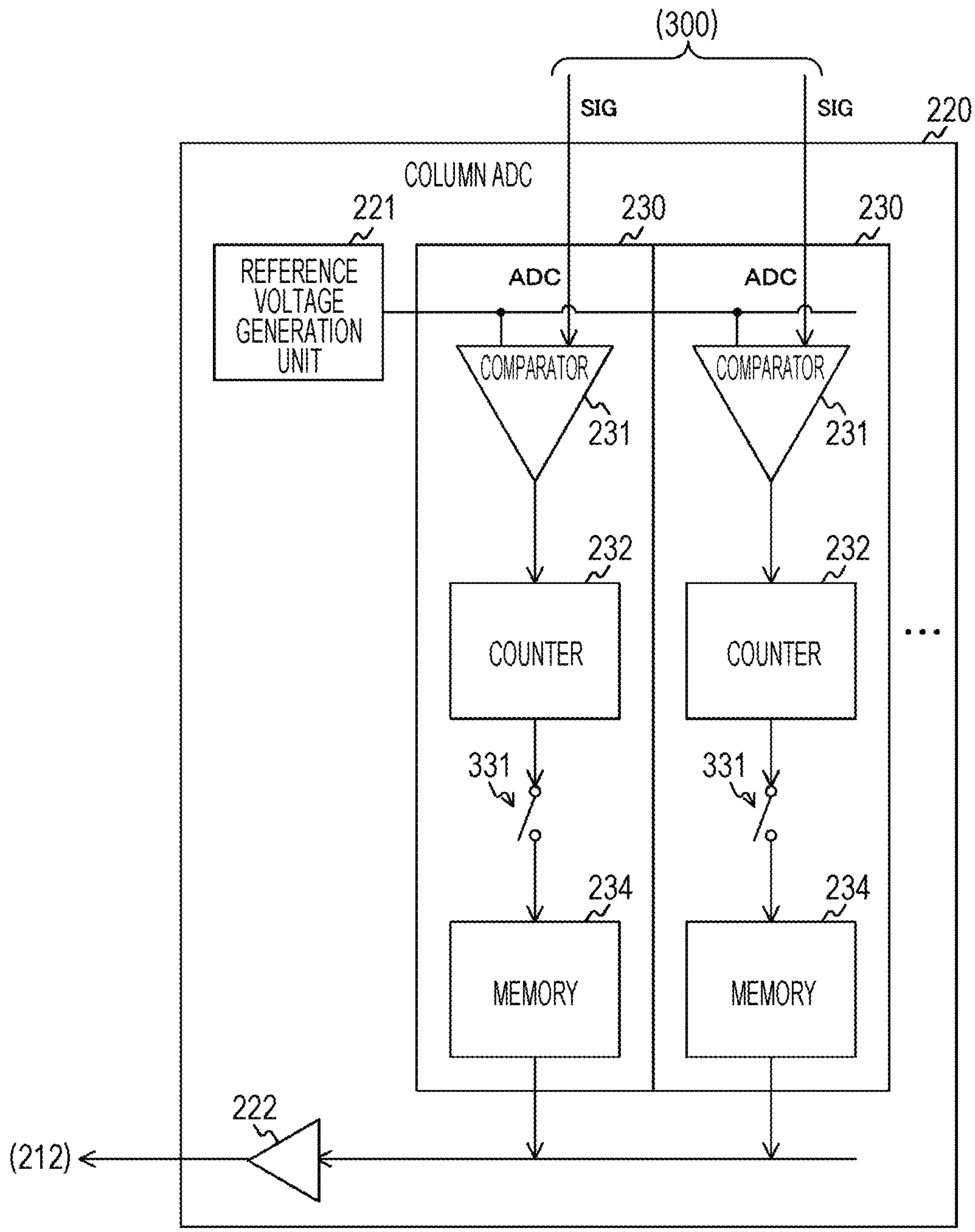

(300)
SIG
SIG
220
COLUMN ADC
221
230
230
REFERENCE VOLTAGE GENERATION UNIT
ADC
ADC
COMPARATOR
COMPARATOR
231
231
232
232
COUNTER
COUNTER
331
331
234
234
MEMORY
MEMORY
222
(212)

FIG. 10

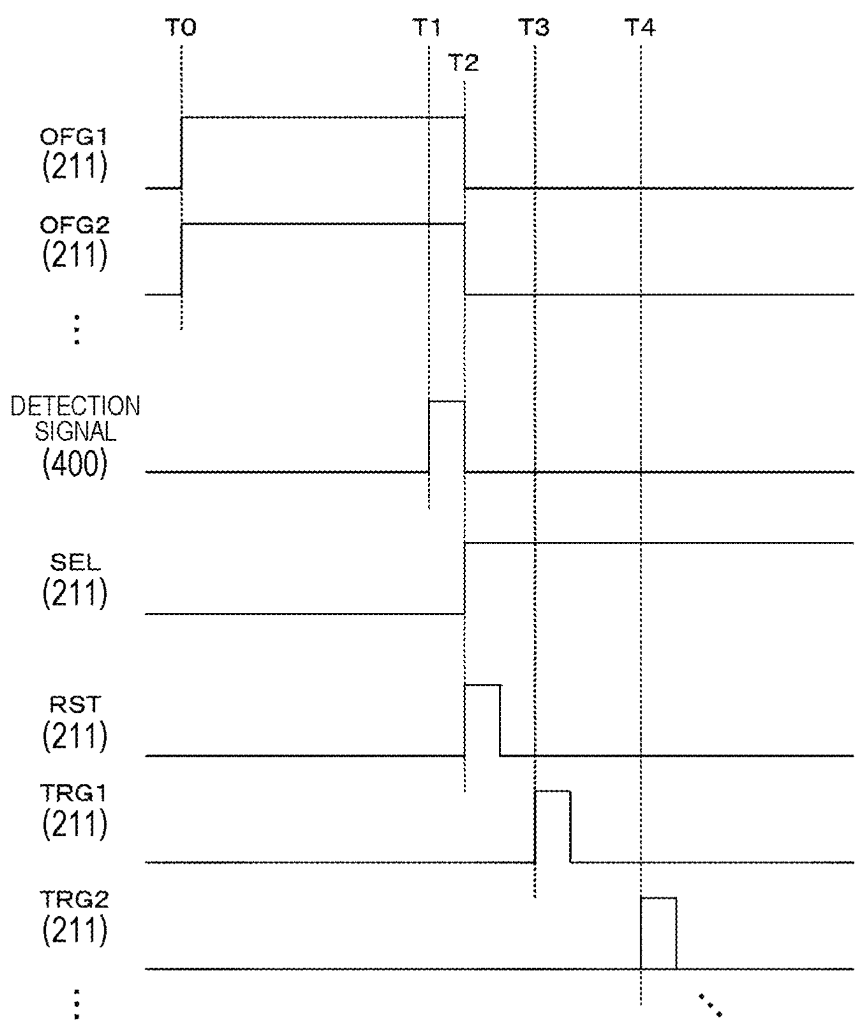

T0
T1
T2
T3
T4
OFG1
(211)
OFG2
(211)
DETECTION
SIGNAL
(400)
SEL
(211)
RST
(211)
TRG1
(211)
TRG2
(211)

FIG. 11

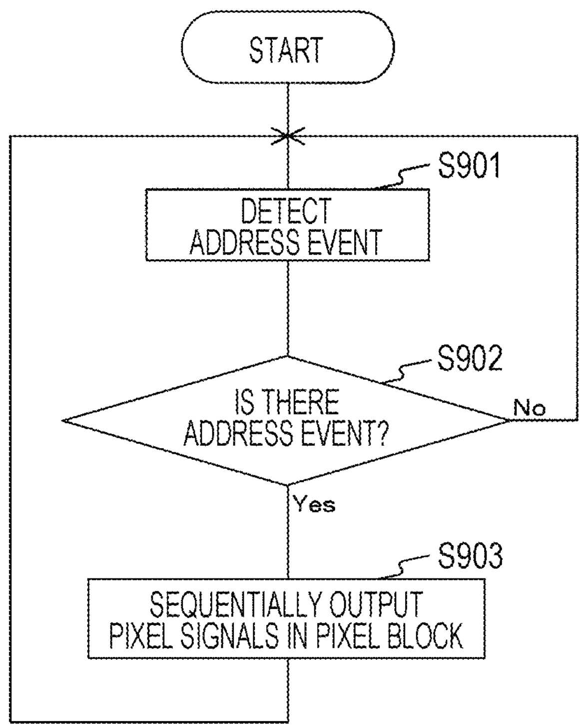
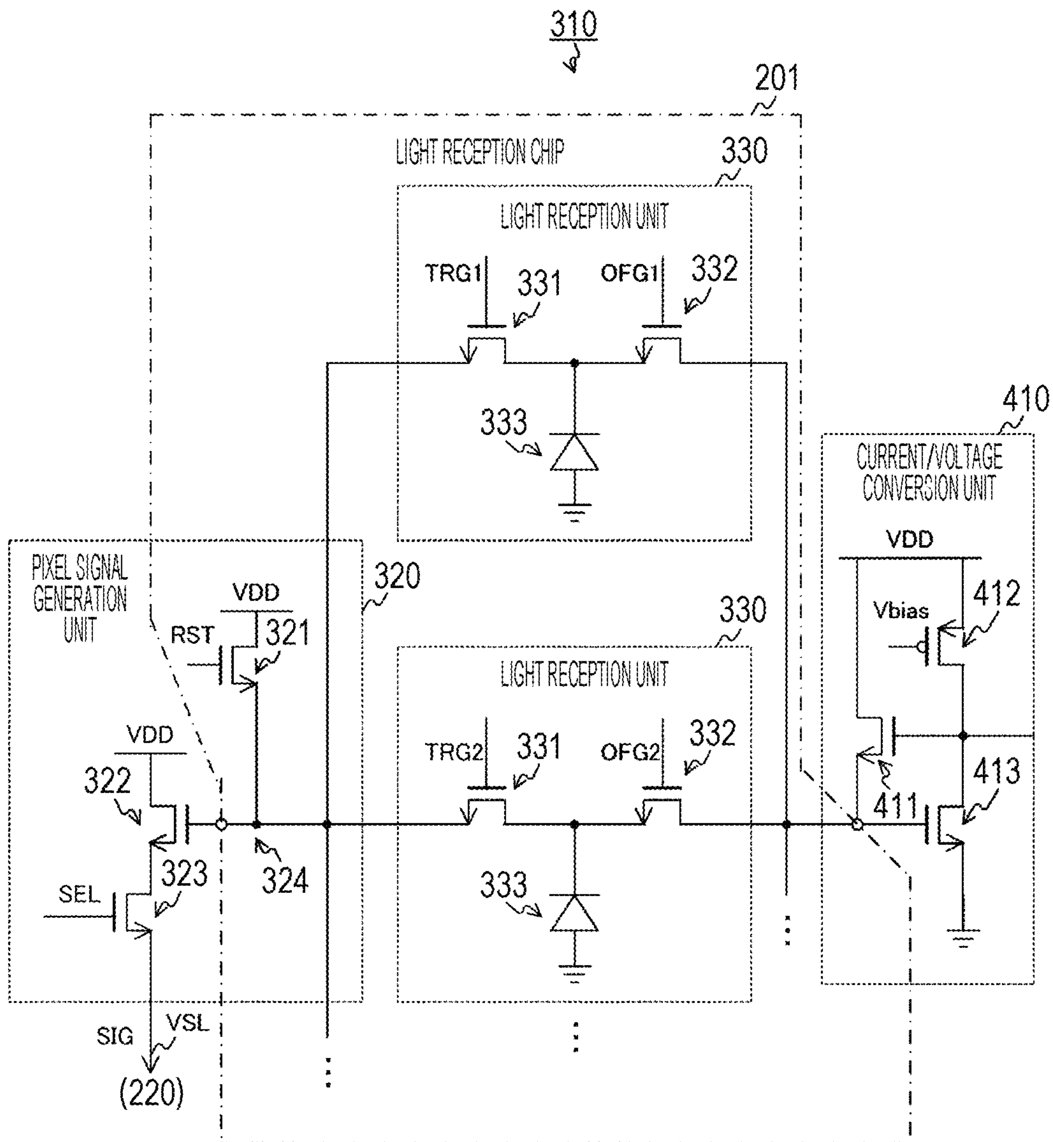

START
S901
DETECT ADDRESS EVENT
S902
IS THERE ADDRESS EVENT?
No
Yes
S903
SEQUENTIALLY OUTPUT PIXEL SIGNALS IN PIXEL BLOCK

FIG. 12



310
201
LIGHT RECEPTION CHIP
330
LIGHT RECEPTION UNIT
TRG1
331
OFG1
332
333
410
CURRENT/VOLTAGE CONVERSION UNIT
VDD
Vbias
412
PIXEL SIGNAL GENERATION UNIT
320
VDD
RST
321
330
LIGHT RECEPTION UNIT
VDD
322
TRG2
331
OFG2
332
413
411
323
324
SEL
333
SIG
VSL
(220)

FIG. 13

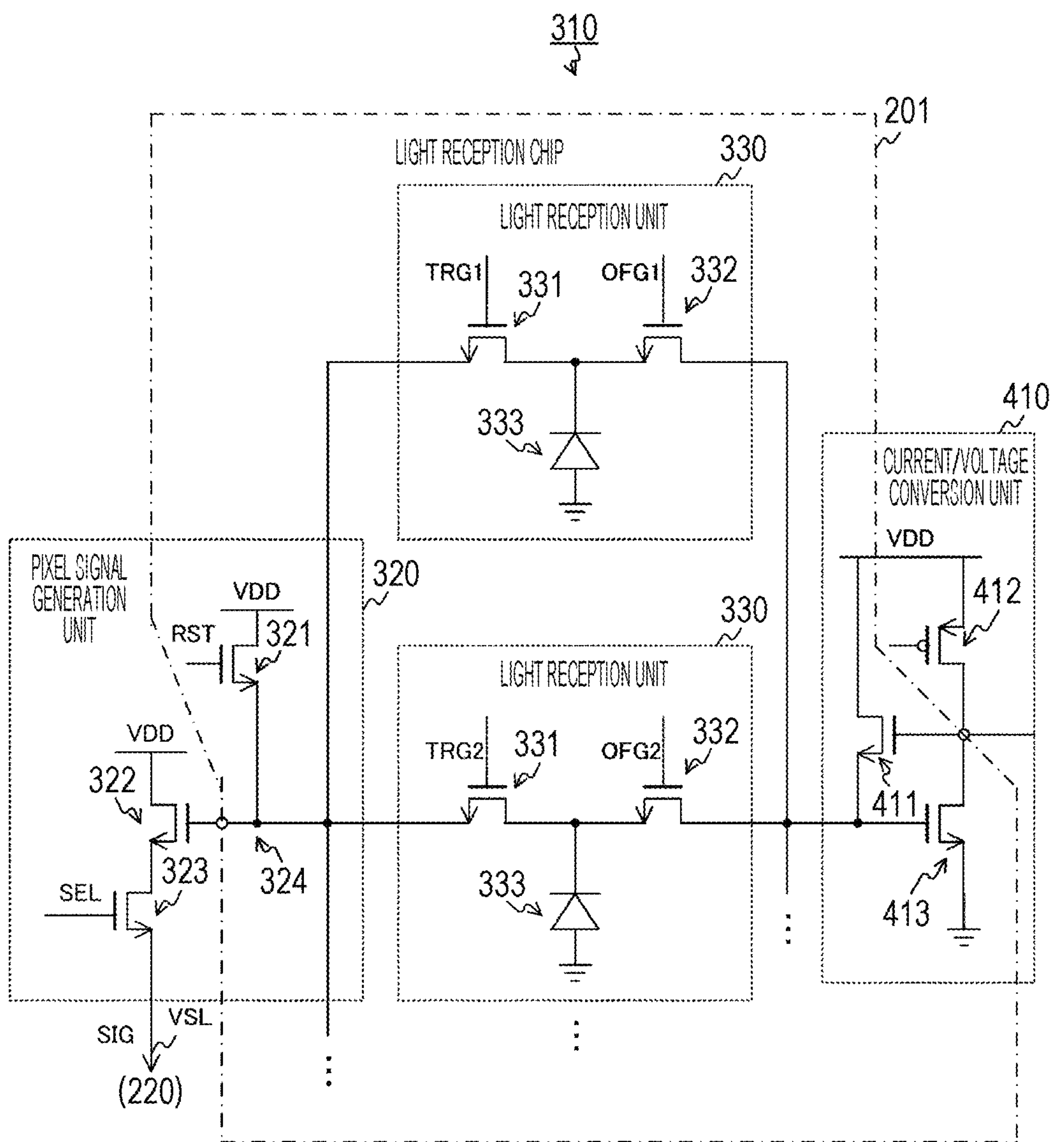

310
201
LIGHT RECEPTION CHIP
330
LIGHT RECEPTION UNIT
TRG1
331
OFG1
332
333
410
CURRENT/VOLTAGE CONVERSION UNIT
VDD
412
411
413
PIXEL SIGNAL GENERATION UNIT
320
VDD
RST
321
330
LIGHT RECEPTION UNIT
TRG2
331
OFG2
332
VDD
322
323
SEL
324
333
SIG
VSL
(220)

FIG. 14

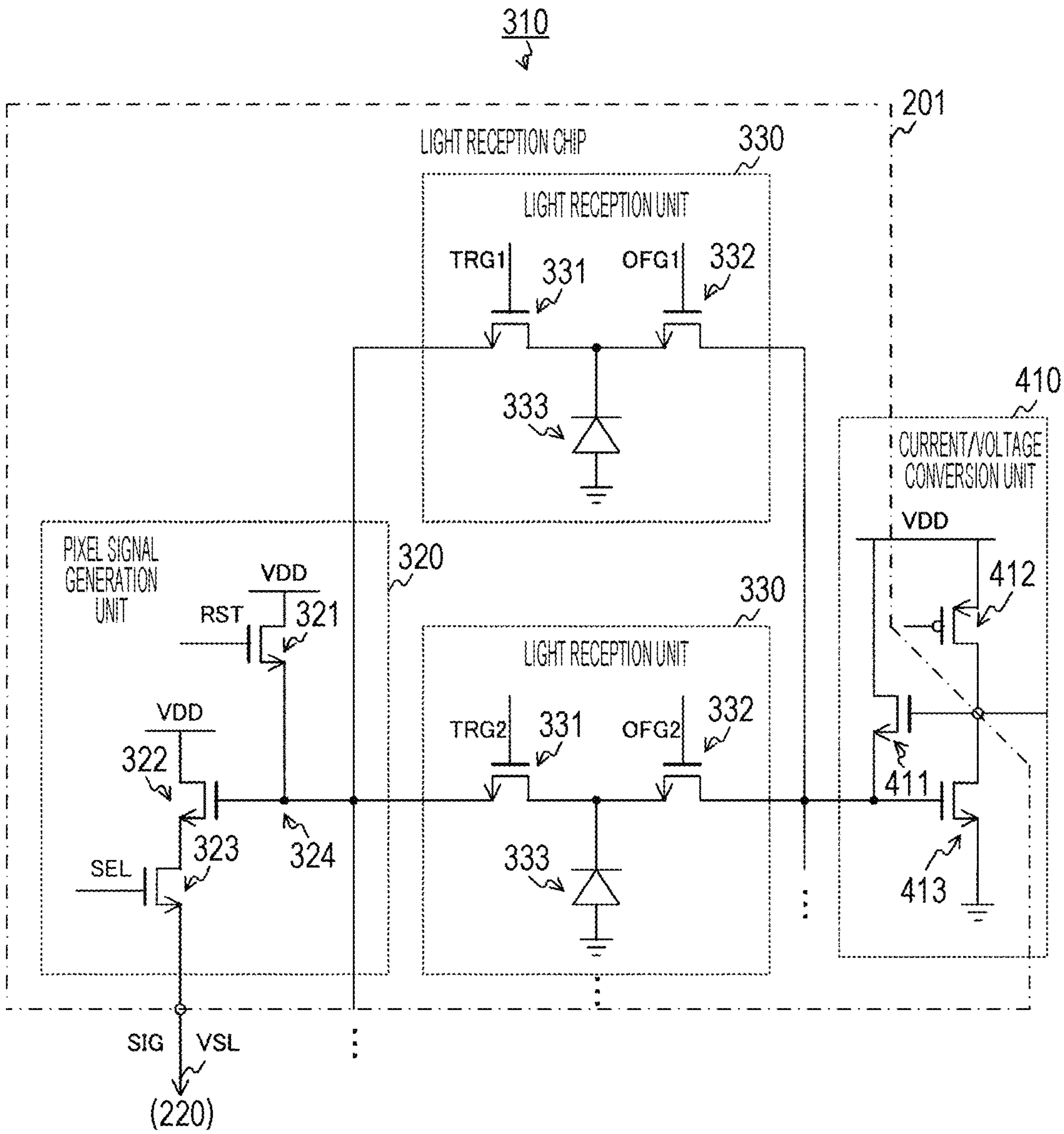

310
201
LIGHT RECEPTION CHIP
330
LIGHT RECEPTION UNIT
TRG1
331
OFG1
332
333
410
CURRENT/VOLTAGE CONVERSION UNIT
VDD
412
PIXEL SIGNAL GENERATION UNIT
320
VDD
RST
321
330
LIGHT RECEPTION UNIT
TRG2
331
OFG2
332
VDD
322
411
323
324
SEL
333
413
SIG
VSL
(220)

FIG. 15

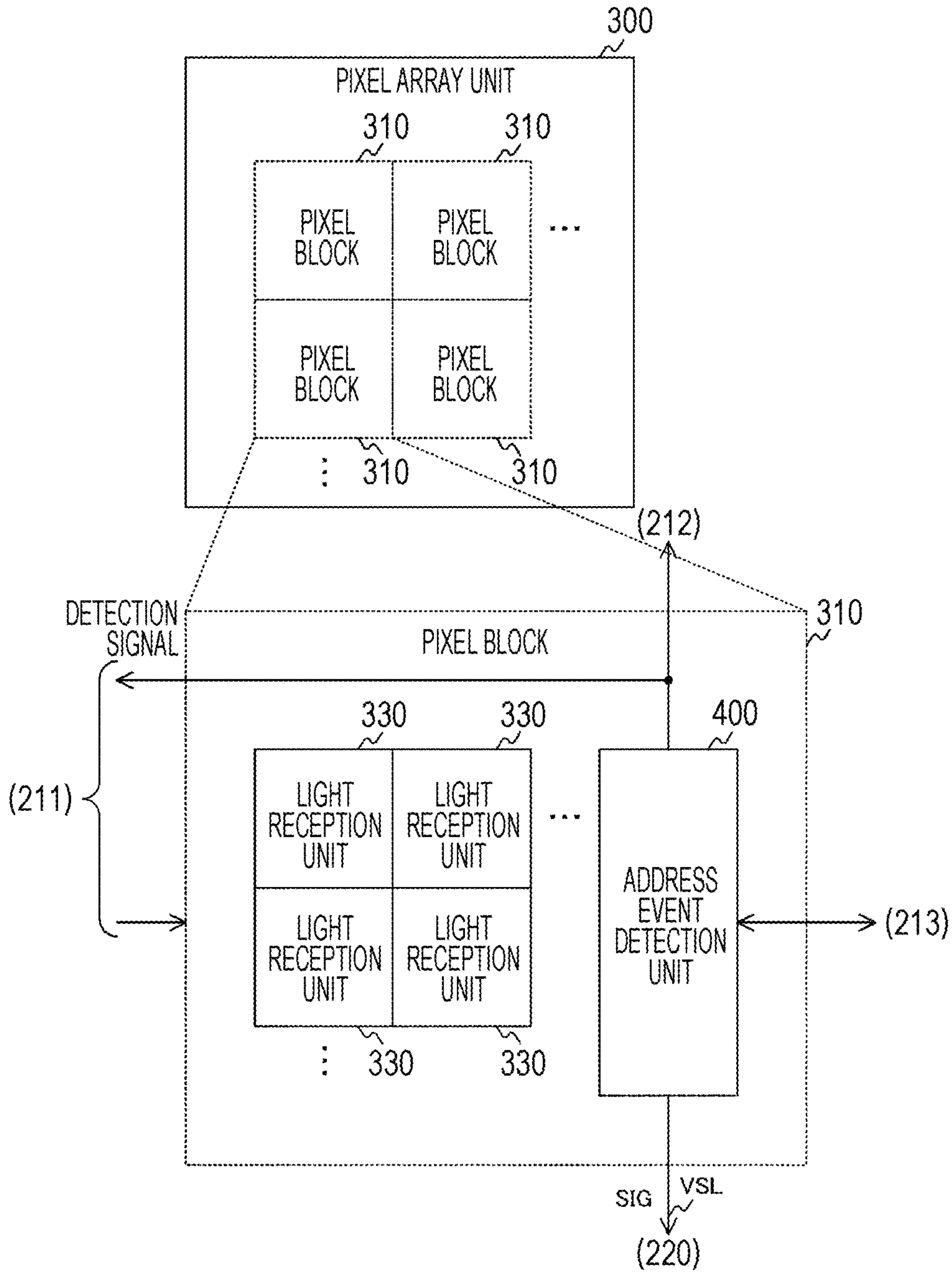

300
PIXEL ARRAY UNIT
310
310
PIXEL BLOCK
PIXEL BLOCK
PIXEL BLOCK
PIXEL BLOCK
310
310
(212)
DETECTION SIGNAL
310
PIXEL BLOCK
330
330
400
(211)
LIGHT RECEPTION UNIT
LIGHT RECEPTION UNIT
LIGHT RECEPTION UNIT
LIGHT RECEPTION UNIT
ADDRESS EVENT DETECTION UNIT
(213)
330
330
SIG
VSL
(220)

FIG. 16

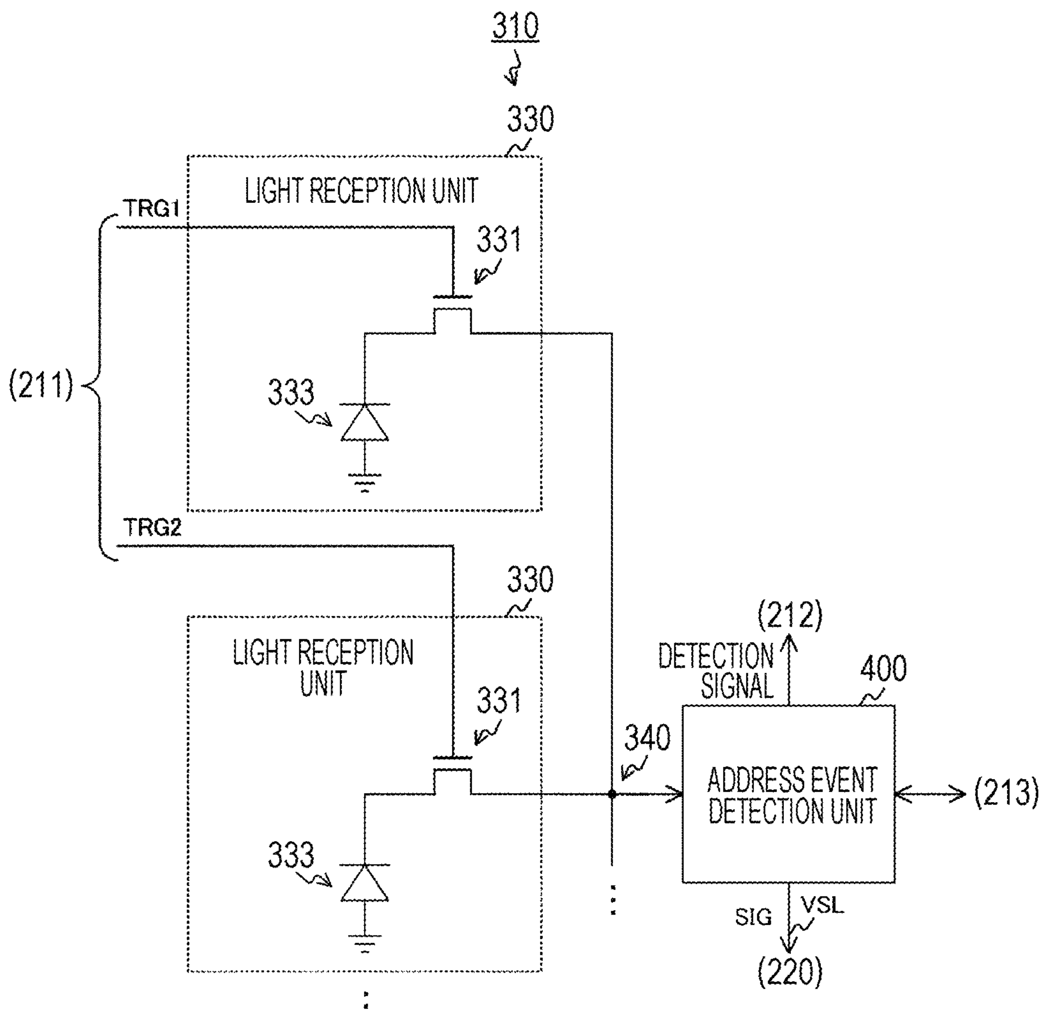

310
330
LIGHT RECEPTION UNIT
TRG1
331
(211)
333
TRG2
330
LIGHT RECEPTION UNIT
331
333
(212)
DETECTION SIGNAL
400
340
ADDRESS EVENT DETECTION UNIT
(213)
SIG
VSL
(220)

FIG. 17

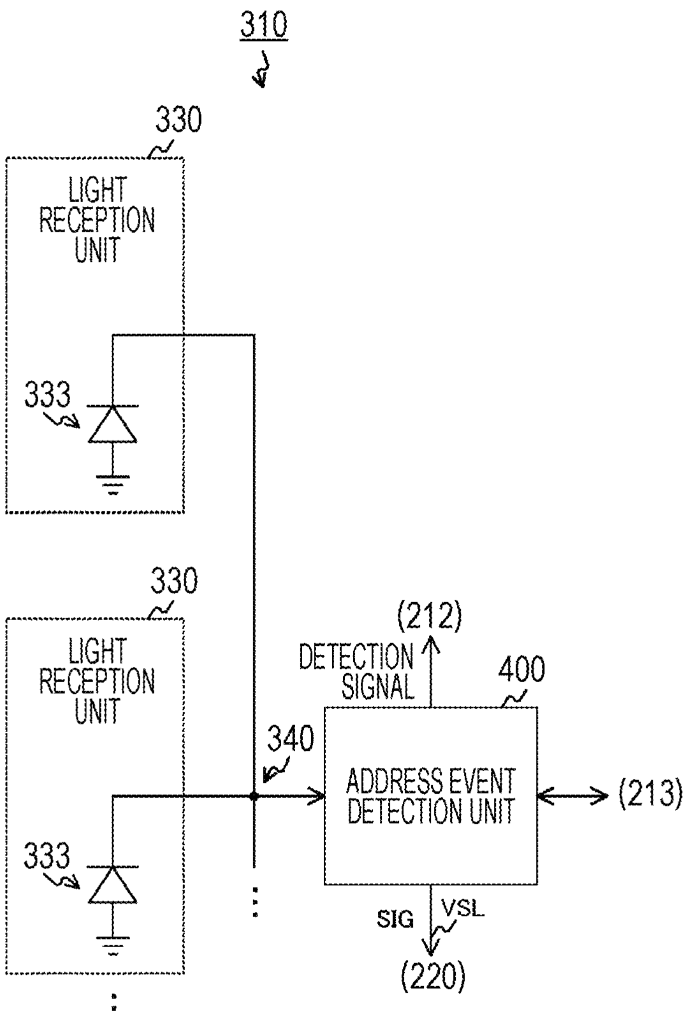
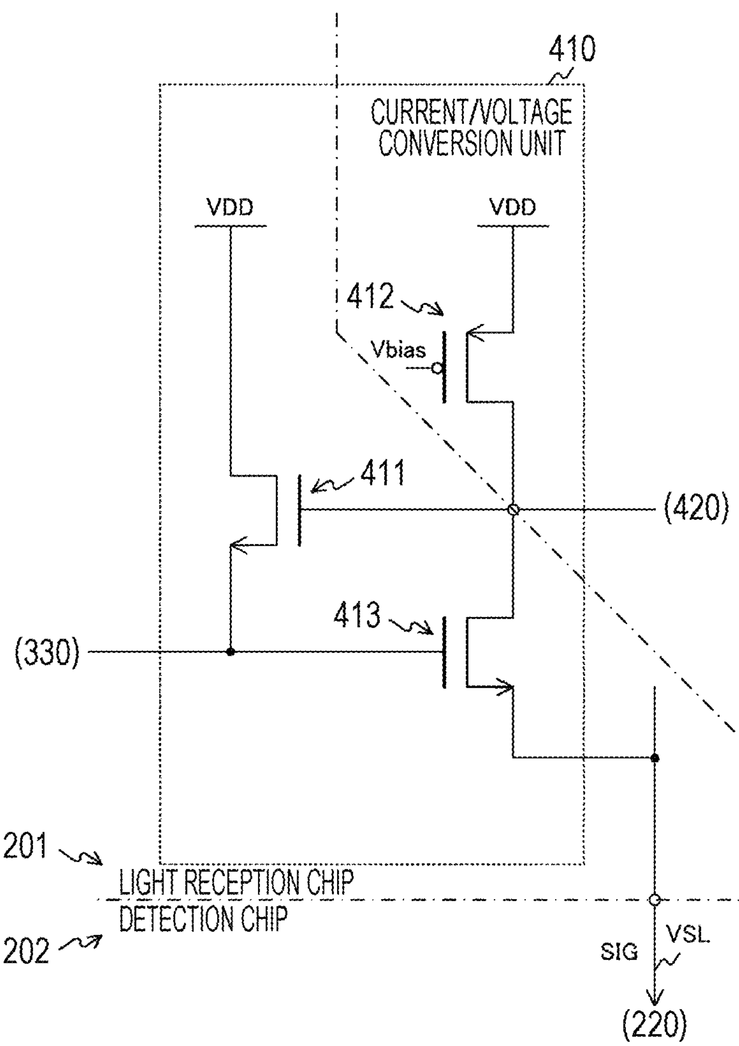

310
330
LIGHT RECEPTION UNIT
333
330
LIGHT RECEPTION UNIT
333
340
(212)
DETECTION SIGNAL
400
ADDRESS EVENT DETECTION UNIT
(213)
SIG
VSL
(220)

FIG. 18



410
CURRENT/VOLTAGE CONVERSION UNIT
VDD
VDD
412
Vbias
411
(420)
413
(330)
201
LIGHT RECEPTION CHIP
DETECTION CHIP
202
SIG
VSL
(220)

FIG. 19

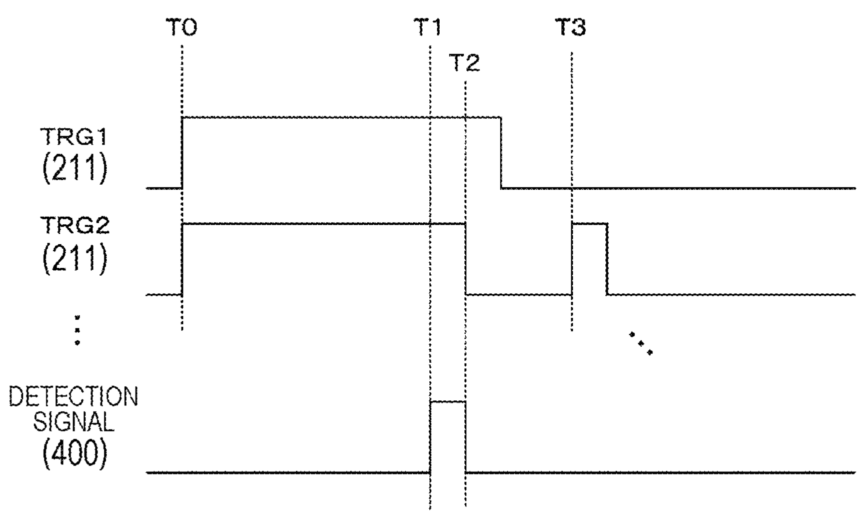
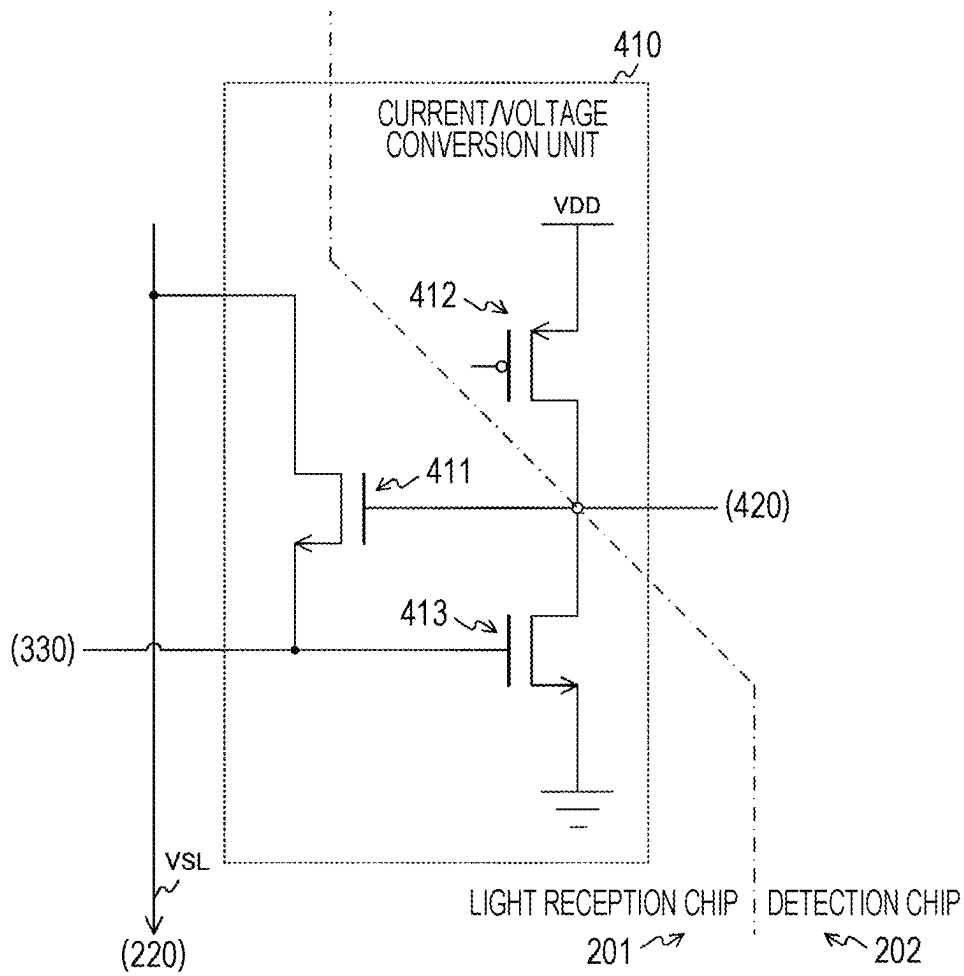


FIG. 20



410
CURRENT/VOLTAGE CONVERSION UNIT
VDD
412
411
(420)
413
(330)
VSL
(220)
LIGHT RECEPTION CHIP
201
DETECTION CHIP
202

FIG. 21

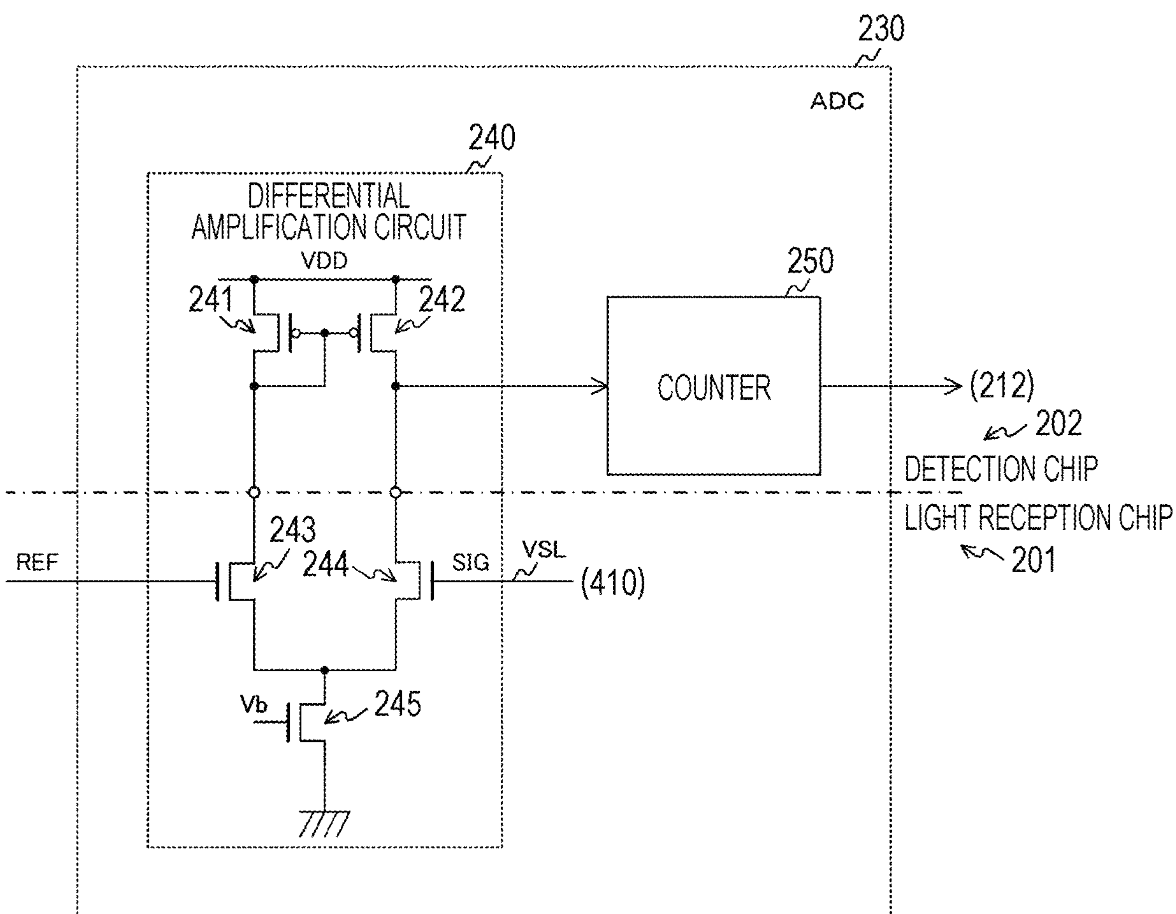

230
ADC
240
DIFFERENTIAL
AMPLIFICATION CIRCUIT
VDD
241
242
250
COUNTER
(212)
202
DETECTION CHIP
LIGHT RECEPTION CHIP
201
243
244
REF
SIG
VSL
(410)
Vb
245

FIG. 22

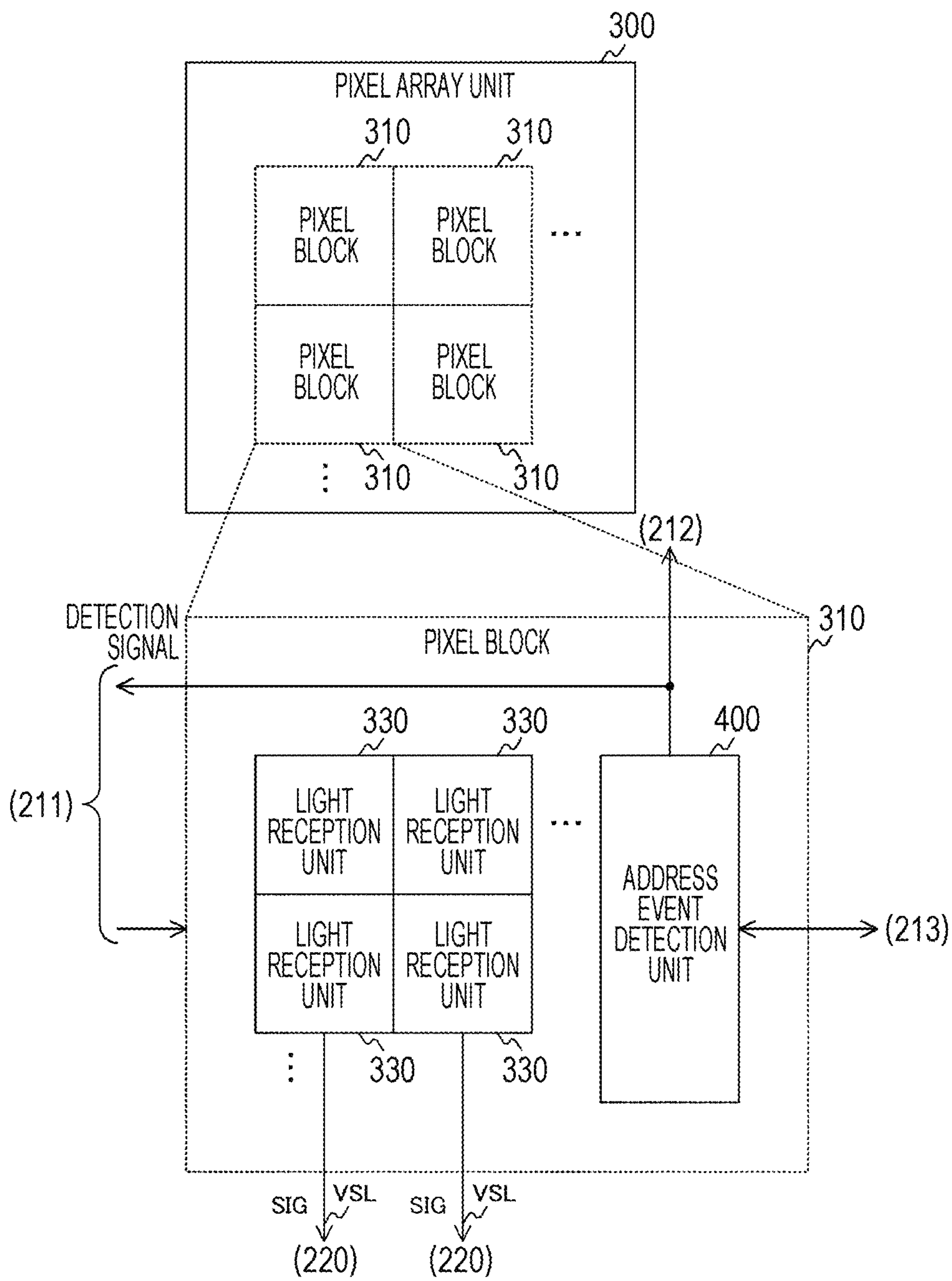
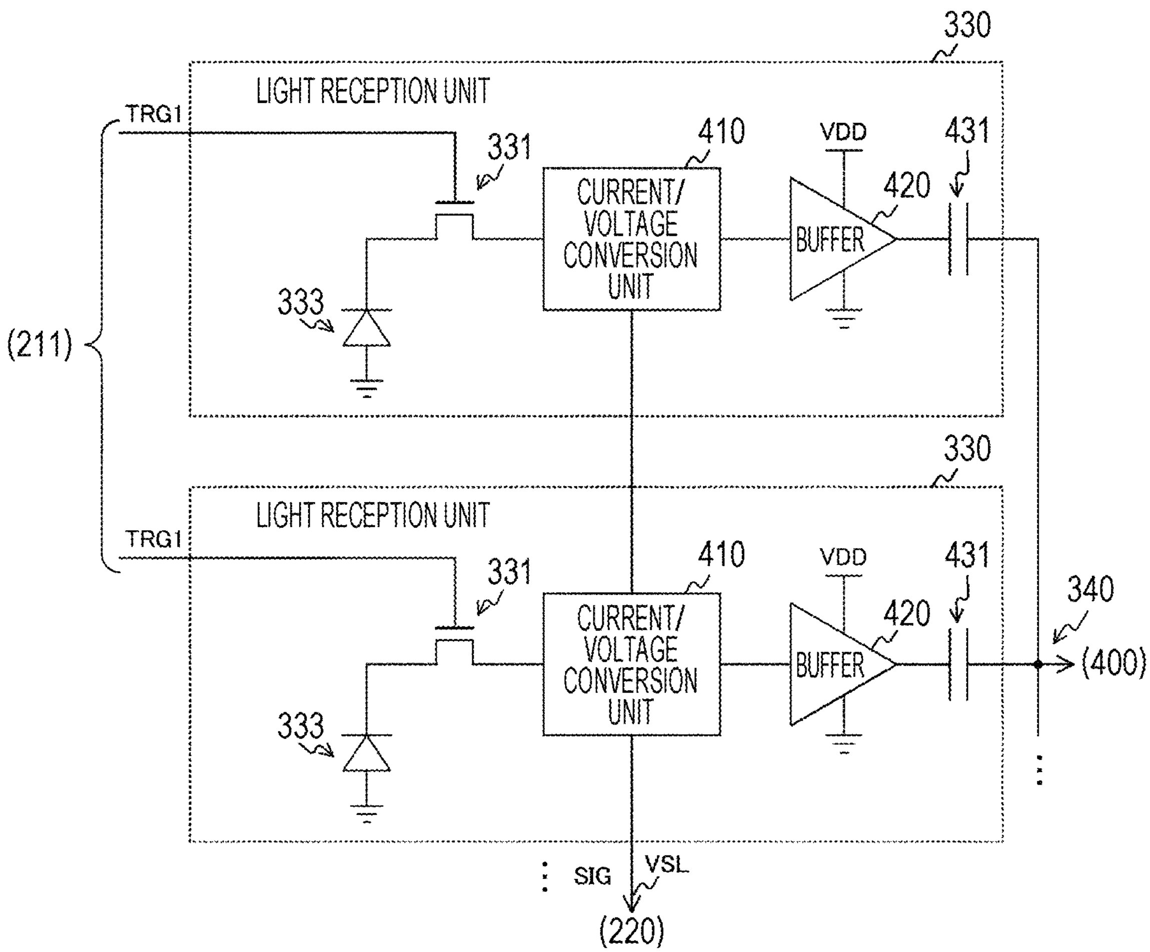

300
PIXEL ARRAY UNIT
310
310
PIXEL BLOCK
PIXEL BLOCK
PIXEL BLOCK
PIXEL BLOCK
310
310
(212)
DETECTION SIGNAL
PIXEL BLOCK
310
330
330
400
(211)
LIGHT RECEPTION UNIT
LIGHT RECEPTION UNIT
LIGHT RECEPTION UNIT
LIGHT RECEPTION UNIT
ADDRESS EVENT DETECTION UNIT
(213)
330
330
SIG
VSL
SIG
VSL
(220)
(220)

FIG. 23



330
LIGHT RECEPTION UNIT
TRG1
331
410
CURRENT/ VOLTAGE CONVERSION UNIT
VDD
431
420
BUFFER
333
(211)
330
LIGHT RECEPTION UNIT
TRG1
331
410
CURRENT/ VOLTAGE CONVERSION UNIT
VDD
431
420
BUFFER
340
(400)
333
SIG
VSL
(220)

FIG. 24

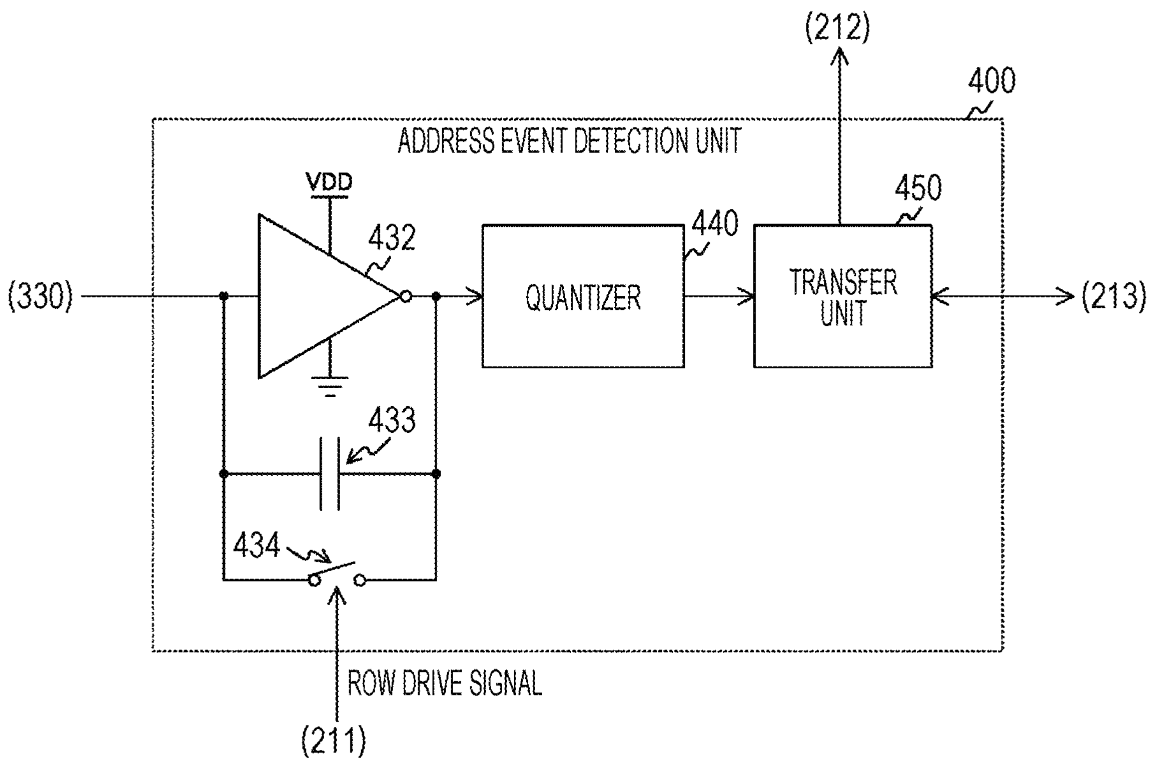
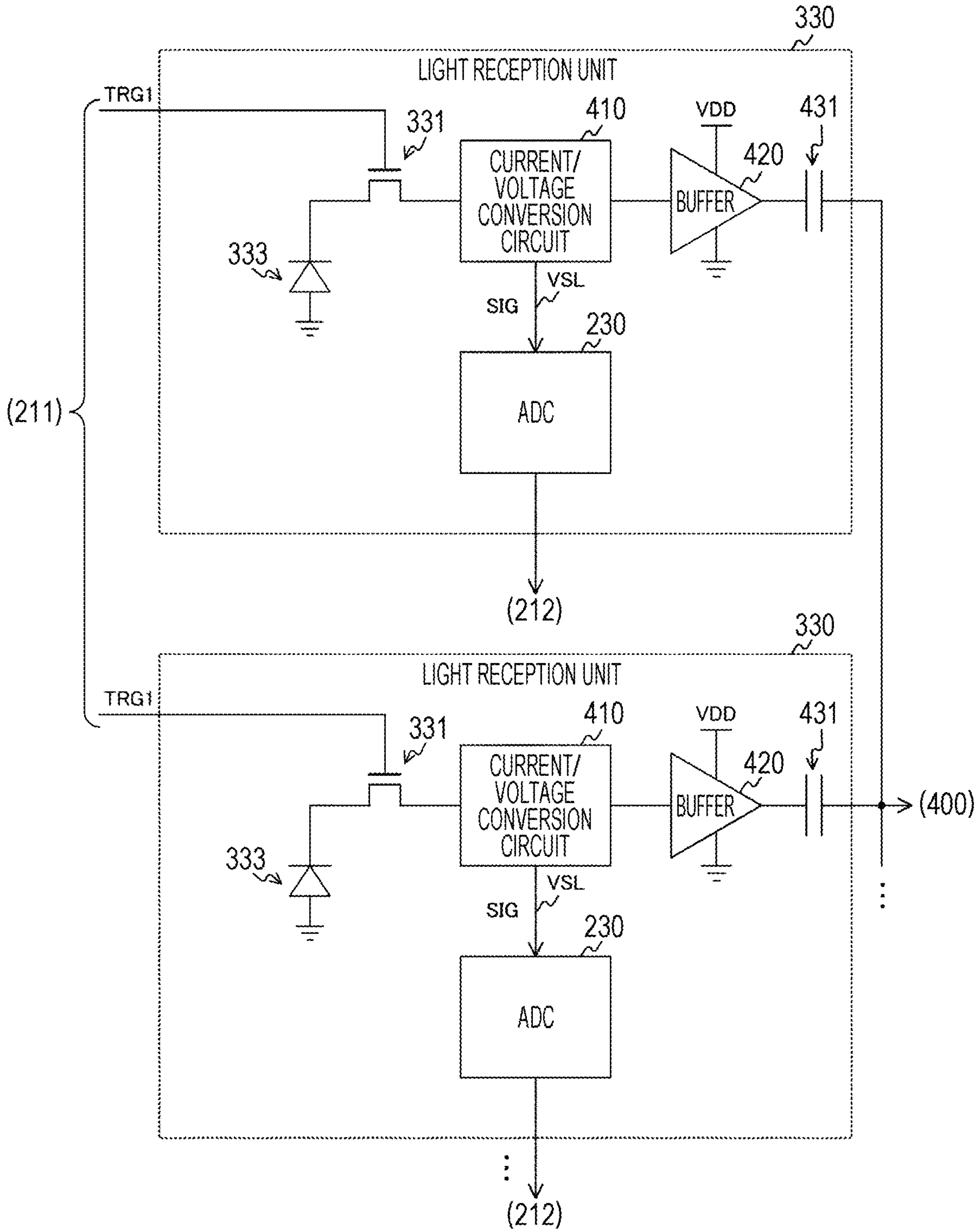
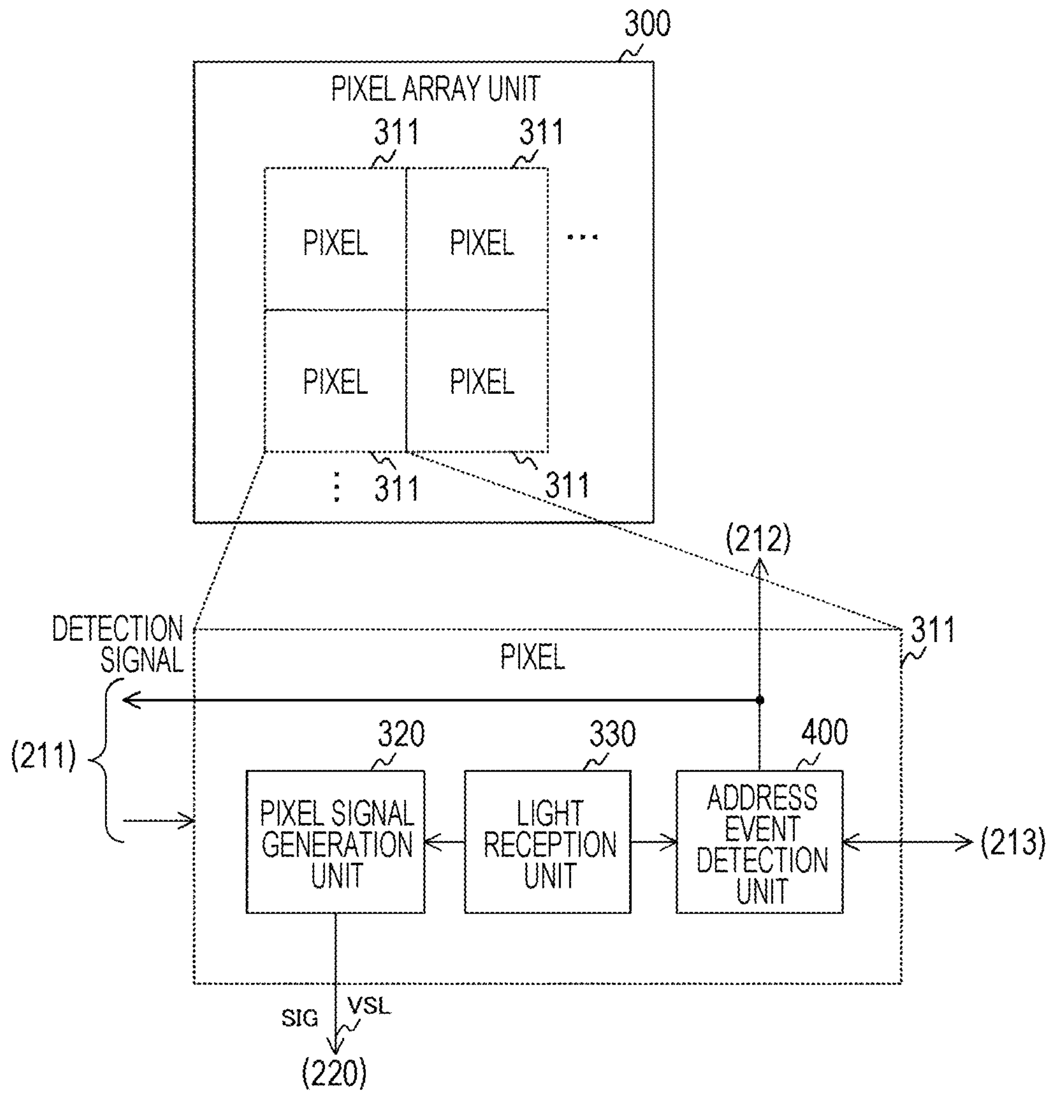

(212)
400
ADDRESS EVENT DETECTION UNIT
VDD
432
440
450
(330)
QUANTIZER
TRANSFER UNIT
(213)
433
434
ROW DRIVE SIGNAL
(211)

FIG. 25



330
LIGHT RECEPTION UNIT
TRG1
331
410
CURRENT/
VOLTAGE
CONVERSION
CIRCUIT
VDD
431
420
BUFFER
333
VSL
SIG
230
ADC
(211)
(212)
330
LIGHT RECEPTION UNIT
TRG1
331
410
CURRENT/
VOLTAGE
CONVERSION
CIRCUIT
VDD
431
420
BUFFER
(400)
333
VSL
SIG
230
ADC
(212)

FIG. 26



300
PIXEL ARRAY UNIT
311
311
PIXEL
PIXEL
PIXEL
PIXEL
311
311
(212)
DETECTION
SIGNAL
PIXEL
311
(211)
320
330
400
PIXEL SIGNAL GENERATION UNIT
LIGHT RECEPTION UNIT
ADDRESS EVENT DETECTION UNIT
(213)
SIG
VSL
(220)

FIG. 27

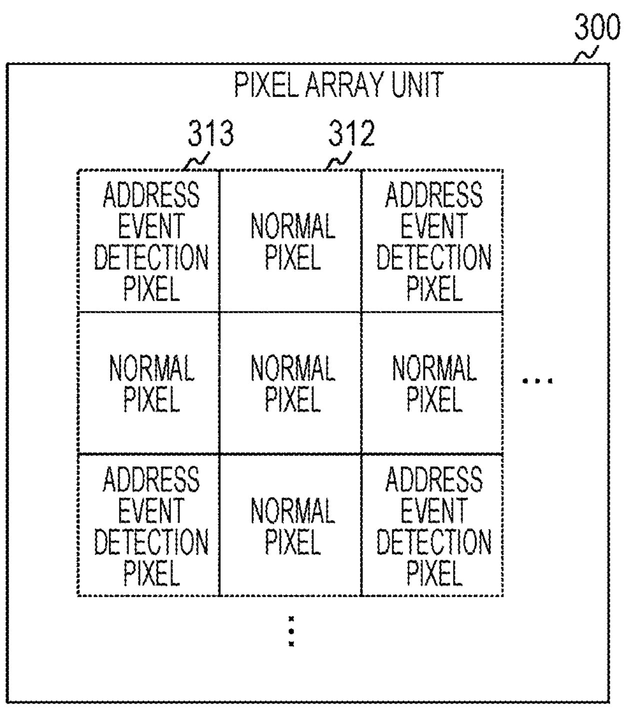
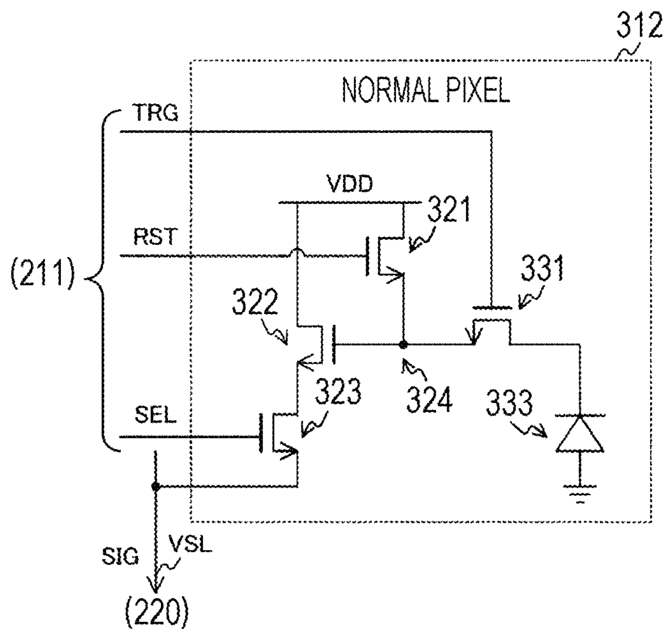
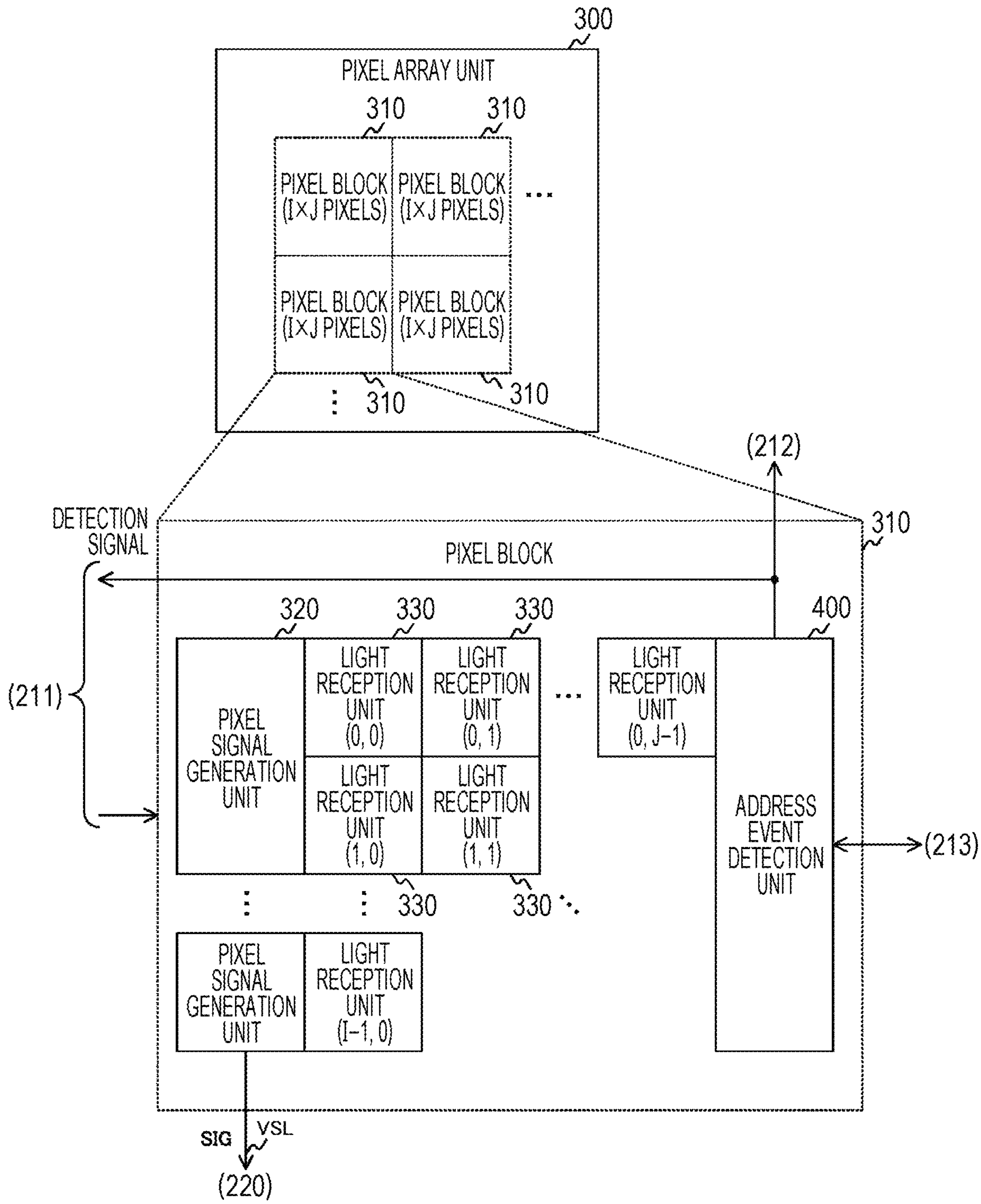

300
PIXEL ARRAY UNIT
313
312
ADDRESS EVENT DETECTION PIXEL
NORMAL PIXEL
ADDRESS EVENT DETECTION PIXEL
NORMAL PIXEL
NORMAL PIXEL
NORMAL PIXEL
ADDRESS EVENT DETECTION PIXEL
NORMAL PIXEL
ADDRESS EVENT DETECTION PIXEL

FIG. 28



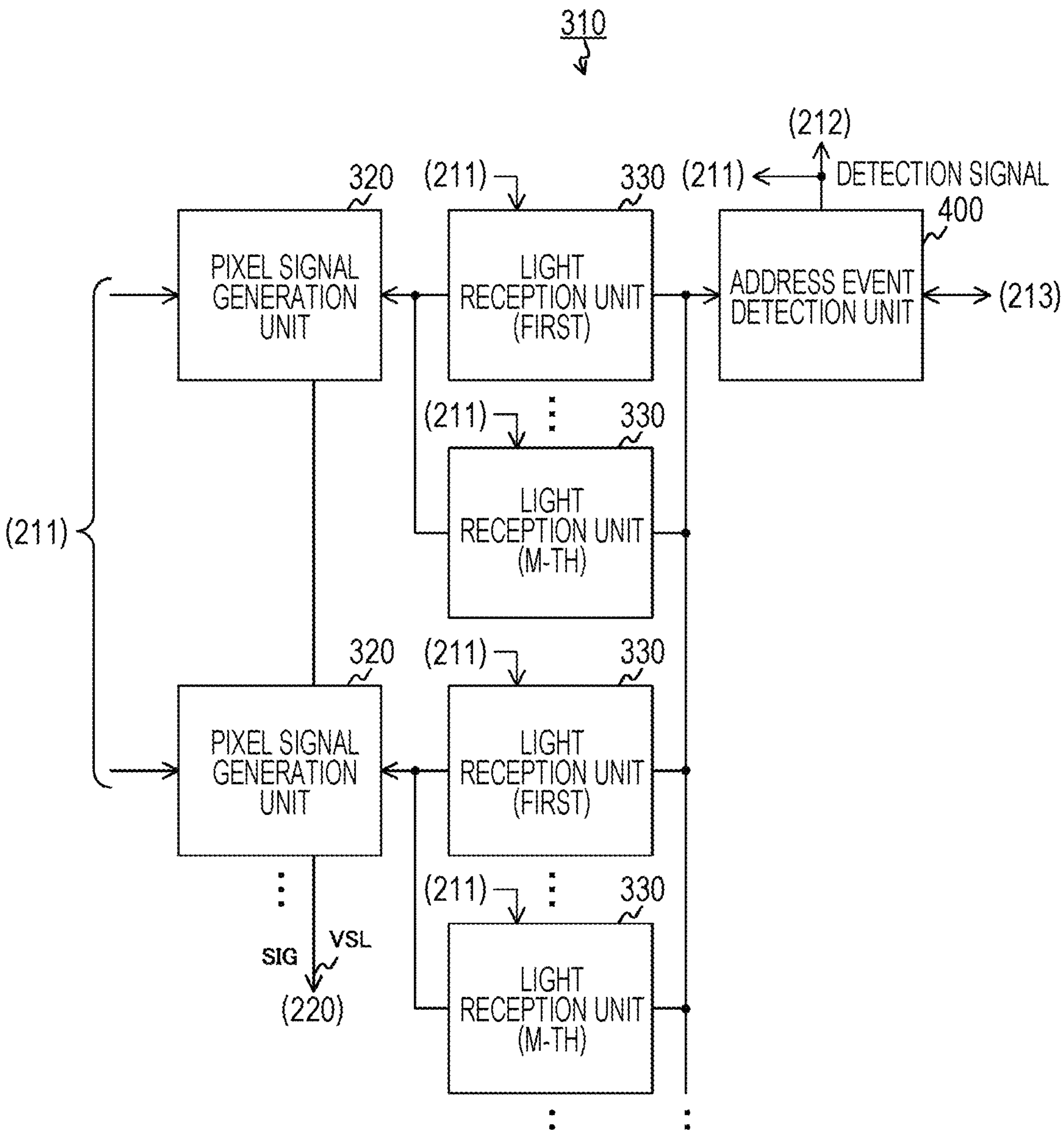
312
NORMAL PIXEL
TRG
VDD
321
RST
331
(211)
322
324
323
333
SEL
SIG
VSL
(220)

FIG. 29



300
PIXEL ARRAY UNIT
310
310
PIXEL BLOCK (I×J PIXELS)
PIXEL BLOCK (I×J PIXELS)
PIXEL BLOCK (I×J PIXELS)
PIXEL BLOCK (I×J PIXELS)
310
310
(212)
310
DETECTION SIGNAL
PIXEL BLOCK
320
330
330
400
(211)
PIXEL SIGNAL GENERATION UNIT
LIGHT RECEPTION UNIT (0, 0)
LIGHT RECEPTION UNIT (0, 1)
LIGHT RECEPTION UNIT (0, J−1)
LIGHT RECEPTION UNIT (1, 0)
LIGHT RECEPTION UNIT (1, 1)
ADDRESS EVENT DETECTION UNIT
(213)
330
330
PIXEL SIGNAL GENERATION UNIT
LIGHT RECEPTION UNIT (I−1, 0)
SIG
VSL
(220)

FIG. 30



310
(212)
320
(211)
330
(211)
DETECTION SIGNAL
400
PIXEL SIGNAL GENERATION UNIT
LIGHT RECEPTION UNIT (FIRST)
ADDRESS EVENT DETECTION UNIT
(213)
(211)
330
LIGHT RECEPTION UNIT (M-TH)
(211)
320
(211)
330
PIXEL SIGNAL GENERATION UNIT
LIGHT RECEPTION UNIT (FIRST)
(211)
330
SIG
VSL
(220)
LIGHT RECEPTION UNIT (M-TH)

FIG. 31

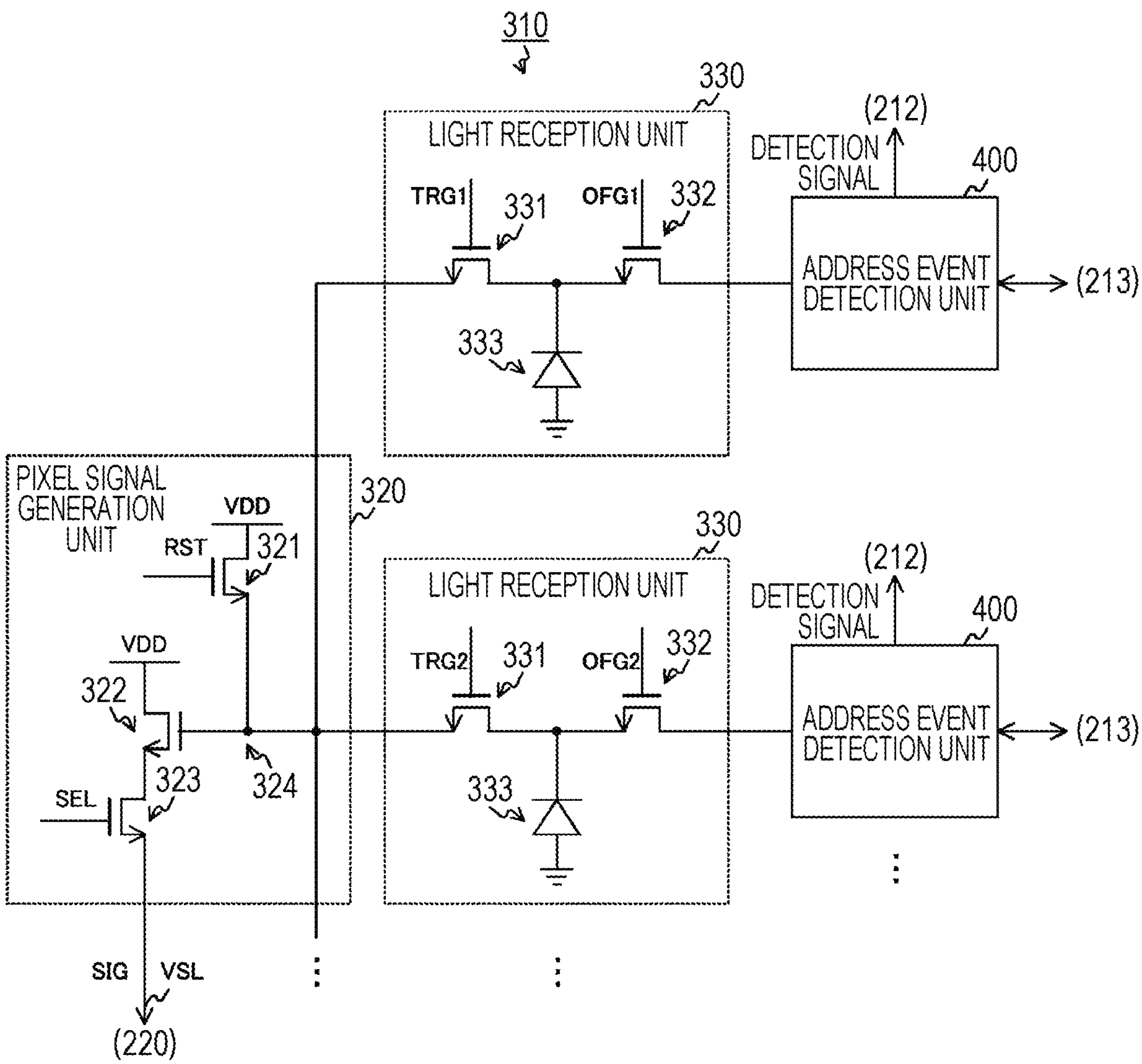

310
330
LIGHT RECEPTION UNIT
TRG1
331
OFG1
332
333
(212)
DETECTION SIGNAL
400
ADDRESS EVENT DETECTION UNIT
(213)
PIXEL SIGNAL GENERATION UNIT
320
VDD
RST
321
VDD
322
323
SEL
324
330
LIGHT RECEPTION UNIT
TRG2
331
OFG2
332
333
(212)
DETECTION SIGNAL
400
ADDRESS EVENT DETECTION UNIT
(213)
SIG
VSL
(220)

FIG. 32

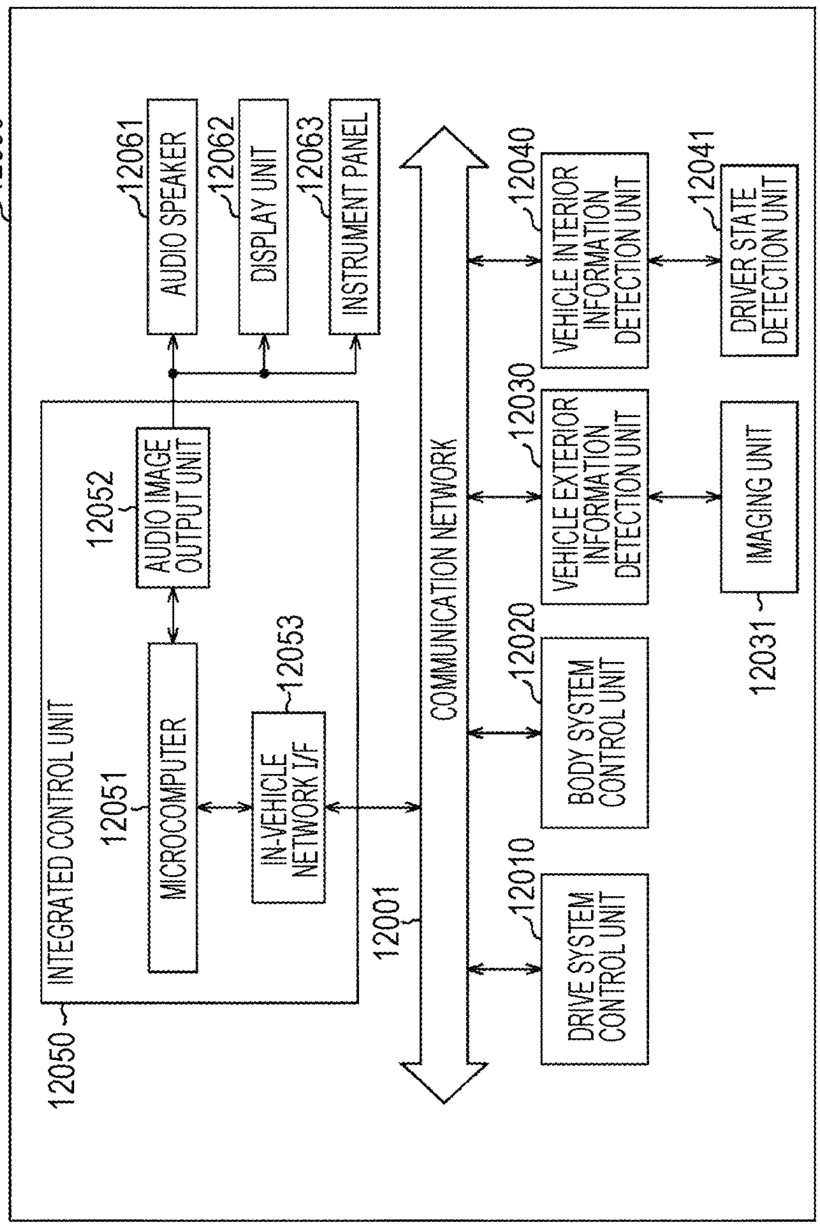
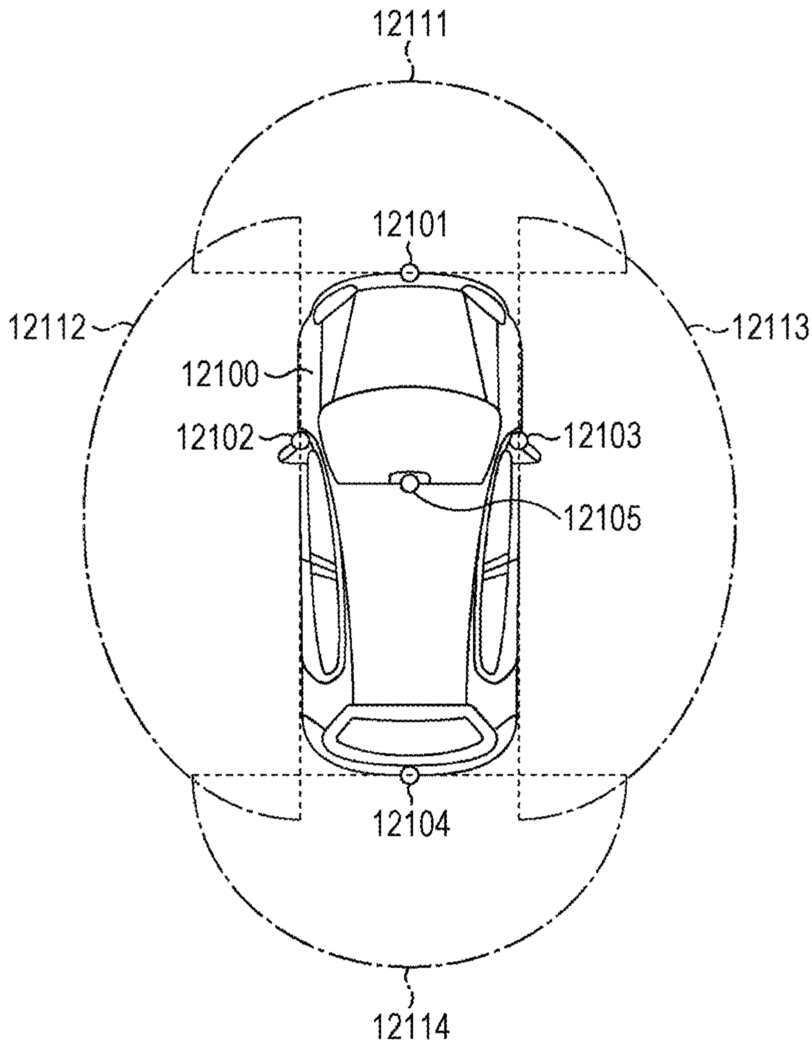

12000
12050
INTEGRATED CONTROL UNIT
12051
MICROCOMPUTER
12052
AUDIO IMAGE OUTPUT UNIT
12053
IN-VEHICLE NETWORK I/F
12061
AUDIO SPEAKER
12062
DISPLAY UNIT
12063
INSTRUMENT PANEL
12001
COMMUNICATION NETWORK
12010
DRIVE SYSTEM CONTROL UNIT
12020
BODY SYSTEM CONTROL UNIT
12030
VEHICLE EXTERIOR INFORMATION DETECTION UNIT
12031
IMAGING UNIT
12040
VEHICLE INTERIOR INFORMATION DETECTION UNIT
12041
DRIVER STATE DETECTION UNIT

FIG. 33


12111
12101
12112
12113
12100
12102
12103
12105
12104
12114

SOLID-STATE IMAGING ELEMENT, IMAGING DEVICE, AND CONTROL METHOD OF SOLID-STATE IMAGING ELEMENT

CROSS REFERENCE TO RELATED APPLICATIONS

The present application is a Continuation of application Ser. No. 18/142,949, filed May 3, 2023, which is a Continuation of application Ser. No. 17/935,343, filed Sep. 26, 2022, now U.S. Pat. No. 11,659,304, issued May 23, 2023, which is a Continuation of application Ser. No. 16/962,783, filed Jul. 16, 2020, now U.S. Pat. No. 11,470,273, issued Oct. 11, 2022, which is a 371 National Stage Entry of International Application No.: PCT/JP2019/001517, filed on Jan. 18, 2019, which claims the benefit of Japanese Priority Patent Application JP 2018-008850 filed Jan. 23, 2018, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

The present technology relates to a solid-state imaging element, an imaging device, and a control method of a solid-state imaging element. More specifically, the present technology relates to a solid-state imaging element that compares an amount of incident light with a threshold, an imaging device, and a control method of a solid-state imaging element.

BACKGROUND ART

Conventionally, a synchronous solid-state imaging element that images image data (frame) in synchronization with a synchronization signal such as a vertical synchronization signal is used in an imaging device and the like. With this general synchronous solid-state imaging element, the image data may be obtained only at every synchronization signal cycle (for example, 1/60 second), so that it is difficult to cope with a case where higher-speed processing is requested in a field regarding traffic, robot and the like. Therefore, an asynchronous solid-state imaging element is proposed in which a detection circuit that detects in real time as an address event that a light amount of the pixel exceeds a threshold for each pixel address is provided for each pixel (refer to, for example, Patent Document 1). The solid-state imaging element that detects the address event for each pixel in this manner is referred to as a dynamic vision sensor (DVS).

CITATION LIST

Patent Document

Patent Document 1: Japanese Unexamined Patent Publication No. 2017-535999

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

The asynchronous solid-state imaging element (that is, DVS) described above may generate data at a much higher speed than that with the synchronous solid-state imaging element to output. For this reason, for example, in a traffic field, processing of recognizing an image of a person or an obstacle may be executed at a high speed to improve safety. However, a detection circuit of the address event has a larger number of elements such as transistors than a pixel circuit in the synchronous type, and there is a problem that, if such circuit is provided for each pixel, a circuit scale increases as compared with the synchronous type.

The present technology is achieved in view of such a situation, and an object thereof is to reduce the circuit scale in the solid-state imaging element that detects the address event.

Solutions to Problems

The present technology is made to solve the above-described problem, and a first aspect thereof is a solid-state imaging element provided with a plurality of photoelectric conversion elements each of which photoelectrically converts incident light to generate a first electric signal, and a detection unit that detects whether or not a change amount of the first electric signal of each of the plurality of photoelectric conversion elements exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection, and a control method thereof. This brings about an effect that the detection result of detecting whether or not the change amount of the electric signals from the plurality of photoelectric conversion elements exceeds the threshold is output.

Furthermore, in the first aspect, a signal supply unit that supplies the first electric signal of each of the plurality of photoelectric conversion elements to a connection node according to a predetermined control signal may be further provided, in which the detection unit may detect whether or not the change amount of the first electric signal supplied to the connection node exceeds the predetermined threshold. This brings about an effect that the detection result of detecting whether or not the change amount of the electric signal supplied to the connection node by the signal supply unit exceeds the threshold is output.

Furthermore, in the first aspect, a pixel signal generation unit that generates a pixel signal according to a second electric signal generated by the photoelectric conversion element may be further provided, in which the signal supply unit may sequentially select the second electric signal of each of the plurality of photoelectric conversion elements to supply to the pixel signal generation unit in a case where the change amount exceeds the predetermined threshold. This brings about an effect that the pixel signals are sequentially generated in a case where the change amount exceeds the threshold.

Furthermore, in the first aspect, the connection node may be connected to N (N is an integer not smaller than 2) of the photoelectric conversion elements, and the pixel signal generation unit may generate a signal of a voltage corresponding to the second electric signal of an element selected according to a selection signal out of M (M is an integer smaller than N) of the photoelectric conversion elements as the pixel signal. This brings about an effect that it is detected that whether or not the change amount of the electric signals from the N photoelectric conversion elements exceeds the threshold, and the pixel signal is generated from a photocurrent of the selected element out of the M photoelectric conversion elements.

Furthermore, in the first aspect, the pixel signal generation unit may be provided with a reset transistor that initializes a floating diffusion layer, an amplification transistor that amplifies a signal of a voltage of the floating diffusion layer, and a selection transistor that outputs the amplified signal as the pixel signal according to a selection signal, in which the detection unit may be provided with a plurality of N-type transistors that converts the first electric signal into a voltage signal of a logarithm of the first electric signal, and a P-type transistor that supplies a constant current to the plurality of N-type transistors. This brings about an effect that the pixel signal generation unit in which the transistors are arranged and the detection unit generate and detect the pixel signal.

Furthermore, in the first aspect, the plurality of photoelectric conversion elements may be arranged on a light reception chip, and the detection unit and the pixel signal generation unit may be arranged on a detection chip stacked on the light reception chip. This brings about an effect of increasing a light receiving area.

Furthermore, in the first aspect, the plurality of photoelectric conversion elements and the reset transistor may be arranged on a light reception chip, and the detection unit, the amplification transistor, and the selection transistor may be arranged on a detection chip stacked on the light reception chip. This brings about an effect that a circuit scale of the detection chip is reduced.

Furthermore, in the first aspect, the plurality of photoelectric conversion elements, the reset transistor, and the plurality of N-type transistors may be arranged on a light reception chip, and the amplification transistor, the selection transistor, and the P-type transistor may be arranged on a detection chip stacked on the light reception chip. This brings about an effect that a circuit scale of the detection chip is reduced.

Furthermore, in the first aspect, the plurality of photoelectric conversion elements, the pixel signal generation unit, and the plurality of N-type transistors may be arranged on a light reception chip, and the P-type transistor may be arranged on a detection chip stacked on the light reception chip. This brings about an effect that a circuit scale of the detection chip is reduced.

Furthermore, in the first aspect, a signal supply unit that supplies the first electric signal of each of the plurality of photoelectric conversion elements to a connection node according to a predetermined control signal may be further provided, in which the detection unit may further output a pixel signal corresponding to the first electric signal, the signal supply unit may sequentially select the first electric signal of each of the plurality of photoelectric conversion elements to supply to the connection node in a case where the change amount exceeds the predetermined threshold, and the detection unit may be provided with first and second N-type transistors that convert the first electric signal into a voltage signal of a logarithm of the first electric signal, and a P-type transistor that supplies a constant current to the first and second N-type transistors. This brings about an effect that the pixel signals are sequentially generated in a case where the change amount exceeds the threshold.

Furthermore, in the first aspect, an analog/digital converter that converts the pixel signal into a digital signal may be further provided, in which the plurality of photoelectric conversion elements, the signal supply unit, and the first and second N-type transistors may be arranged on a light reception chip, and the P-type transistor and at least a part of the analog/digital converter may be arranged on a detection chip stacked on the light reception chip. This brings about an effect that a circuit scale of the detection chip is reduced.

Furthermore, in the first aspect, the analog/digital converter may be provided with a signal side transistor to which the pixel signal is input, a reference side transistor to which a predetermined reference signal is input, a constant current source connected to the signal side transistor and the reference side transistor, and a current mirror circuit that amplifies a difference between the pixel signal and the predetermined reference signal to output, and the plurality of photoelectric conversion elements, the signal supply unit, the first and second N-type transistors, the signal side transistor, the reference side transistor, and the constant current source may be arranged on a light reception chip, and the P-type transistor and the current mirror circuit may be arranged on a detection chip stacked on the light reception chip. This brings about an effect that a circuit scale of the detection chip is reduced.

Furthermore, in the first aspect, a connection node connected to the photoelectric conversion element and the detection unit, and for each of the plurality of photoelectric conversion elements, a current/voltage conversion unit that converts a photocurrent into a voltage signal of a logarithm of the photocurrent, a buffer that corrects the voltage signal to output, a capacitor inserted between the buffer and the connection node, and a signal processing unit that supplies an electric signal of each of the plurality of photoelectric conversion elements to the connection node through the current/voltage conversion unit, the buffer, and the capacitor according to a predetermined control signal may be further provided, in which the electric signal may include the photocurrent and the voltage signal. This brings about an effect that the voltage signal of the logarithm of the photocurrent is supplied to the connection node.

Furthermore, in the first aspect, an analog/digital converter that converts a pixel signal into a digital signal may be further provided, in which each of a predetermined number of current/voltage conversion units arranged in a predetermined direction may further generate a signal of a voltage corresponding to the photocurrent as the pixel signal, and output the pixel signal to the analog/digital converter. This brings about an effect that the pixel signals of a predetermined number of pixels are sequentially converted into digital signals.

Furthermore, in the first aspect, an analog/digital converter that converts a pixel signal into a digital signal for each of the plurality of photoelectric conversion elements may be further provided, in which each of current/voltage conversion units may further generate a signal of a voltage corresponding to the photocurrent as the pixel signal, and output the pixel signal to the analog/digital converter. This brings about an effect that the pixel signal is converted into the digital signal for each pixel.

Furthermore, a second aspect of the present technology is a solid-state imaging element provided with a photoelectric conversion element that photoelectrically converts incident light to generate an electric signal, a signal supply unit that supplies the electric signal to either a connection node or a floating diffusion layer according to a predetermined control signal, a detection unit that detects whether or not a change amount of the electric signal supplied to the connection node exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection, and a pixel signal generation unit that generates a voltage signal corresponding to the electric signal supplied to the floating diffusion layer as a pixel signal. Therefore, the pixel signal is generated for each pixel, and it is detected whether or not the change amount of the electric signal exceeds the threshold.

Furthermore, in the second aspect, the signal supply unit may include a first transistor that supplies the electric signal to the connection node according to a predetermined control signal, and a second transistor that supplies the electric signal to a floating diffusion layer according to a predetermined control signal, the pixel signal generation unit may be arranged in each of a plurality of pixels, and the first transistor and the detection unit may be arranged in a pixel being a detection target out of the plurality of pixels. This brings about an effect of reducing a circuit scale.

Furthermore, a third aspect of the present technology is an imaging device provided with a plurality of photoelectric conversion elements each of which photoelectrically converts incident light to generate an electric signal, a signal supply unit that supplies the electric signal of each of the plurality of photoelectric conversion elements to a connection node according to a predetermined control signal, a detection unit that detects whether or not a change amount of the electric signal supplied to the connection node exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection, and a recording unit that records the detection signal. This brings about an effect that the detection result of detecting whether or not the change amount of the electric signals from the plurality of photoelectric conversion elements exceeds the threshold is recorded.

Furthermore, a fourth aspect of the present technology is a solid-state imaging element provided with a first photoelectric conversion element that generates a first electric signal, a second photoelectric conversion element that generates a second electric signal, a detection unit that detects whether or not at least any one of a change amount of the first electric signal or a change amount of the second electric signal exceeds a predetermined threshold to output a detection signal indicating a result of the detection, and a connection node connected to the first photoelectric conversion element, the second photoelectric conversion element, and the detection unit. This brings about an effect that the detection result of whether or not the change amount of the electric signal from any one of the plurality of photoelectric conversion elements exceeds the threshold is output.

Furthermore, in the fourth aspect, a first transistor that supplies the first electric signal to the connection node according to a first control signal, and a second transistor that supplies the second electric signal to the connection node according to a second control signal may be further provided, in which the detection unit may detect whether or not a change amount of either the first or second electric signal supplied to the connection node exceeds the predetermined threshold. This brings about an effect that the detection result of detecting whether or not the change amount of the electric signal supplied to the connection node by the signal supply unit exceeds the threshold is output.

Furthermore, in the fourth aspect, a pixel signal generating unit that generates a first pixel signal according to a third electric signal generated by the first photoelectric conversion element and generates a second pixel signal according to a fourth electric signal generated by the second photoelectric conversion element, a third transistor connected to the first photoelectric conversion element and the pixel signal generating unit, and a fourth transistor connected to the second photoelectric conversion element and the pixel signal generating unit may be further provided, in which the third transistor may supply the third electric signal to the pixel signal generating unit in a case where the change amount of the first electric signal exceeds the predetermined threshold, and the fourth transistor may supply the fourth electric signal to the pixel signal generating unit in a case where the change amount of the second electric signal exceeds the predetermined threshold. This brings about an effect that the pixel signals are sequentially generated in a case where the change amount exceeds the threshold.

Furthermore, in the fourth aspect, the pixel signal generating unit may include a first pixel signal generation unit that generates a first pixel signal according to the third electric signal generated by the first photoelectric conversion element, and a second pixel signal generation unit that generates a second pixel signal according to the fourth electric signal generated by the second photoelectric conversion element, in which the third transistor may supply the third electric signal to the first pixel signal generation unit in a case where the change amount of the first electric signal exceeds the predetermined threshold, and the fourth transistor may supply the fourth electric signal to the second pixel signal generation unit in a case where the change amount of the second electric signal exceeds the predetermined threshold. This brings about an effect that the pixel signals are sequentially generated in a case where the change amount exceeds the threshold.

Furthermore, in the fourth aspect, a third photoelectric conversion element that generates a fifth electric signal and a sixth electric signal, a fifth transistor that supplies an electric signal of the fifth photoelectric conversion element to the connection node according to a third control signal, and a second pixel signal generating unit that generates a third pixel signal according to the sixth electric signal may be further provided, in which the sixth transistor may supply the sixth electric signal to the second pixel signal generating unit in a case where a change amount of the fifth electric signal exceeds the predetermined threshold. This brings about an effect that the pixel signals are sequentially generated by the plurality of pixel signal generating units.

Furthermore, in the fourth aspect, the pixel signal generating unit may be provided with a reset transistor that initializes a floating diffusion layer, an amplification transistor that amplifies a signal of a voltage of the floating diffusion layer, and a selection transistor that outputs the amplified signal as the first or second pixel signal according to a selection signal, and the detection unit may be provided with a plurality of N-type transistors that converts a photocurrent into a voltage signal of a logarithm of the photocurrent, and a P-type transistor that supplies a constant current to the plurality of N-type transistors. This brings about an effect that the pixel signals are sequentially generated in a case where the change amount exceeds the threshold.

Furthermore, in the fourth aspect, the first electric signal may include a first photocurrent, and the second electric signal may include a second photocurrent, a connection node connected to the first photoelectric conversion element, the second photoelectric conversion element, and the detection unit, a first current/voltage conversion unit that converts at least one of the first photocurrent or the second photocurrent into a voltage signal of a logarithm of the photocurrent, a buffer that corrects the voltage signal to output, a capacitor inserted between the buffer and the connection node, and a signal processing unit that supplies at least one of the first electric signal or the second electric signal to the connection node through the current/voltage conversion unit, the buffer, and the capacitor according to a predetermined control signal may be further provided, in which the first photoelectric conversion element may generate the first photocurrent, and the second photoelectric conversion element may generate the second photocurrent. This brings about an effect that the voltage signal of the logarithm of the photocurrent is supplied to the connection node.

Furthermore, in the fourth aspect, an analog/digital converter connected to the first current/voltage conversion unit and the second current/voltage conversion unit may be further provided, in which the first current/voltage conversion unit may further generate a signal of a voltage corresponding to the first photocurrent as a first pixel signal, and output the first pixel signal to the analog/digital converter, and the second current/voltage conversion unit may further generate a signal of a voltage corresponding to the second photocurrent as a second pixel signal, and output the second pixel signal to the analog/digital converter. This brings about an effect that the pixel signals of a predetermined number of pixels are sequentially converted into digital signals.

Furthermore, in the fourth aspect, a first analog/digital converter that converts a first pixel signal into a first digital signal, and a second analog/digital converter that converts a second pixel signal into a second digital signal may be further provided, in which the first current/voltage conversion unit may further generate a signal of a voltage corresponding to the first photocurrent as the first pixel signal, and outputs the first pixel signal to the first analog/digital converter, and the second current/voltage conversion unit may further generate a signal of a voltage corresponding to the second photocurrent as the second pixel signal, and output the second pixel signal to the second analog/digital converter. This brings about an effect that the pixel signals of a predetermined number of pixels are sequentially converted into digital signals.

Furthermore, a fifth aspect of the present technology is a solid-state imaging element provided with a first photoelectric conversion element that photoelectrically converts incident light to generate a first electric signal and a second electric signal, a first signal supply unit that supplies the first electric signal to a connection node according to a first control signal, a second signal supply unit that supplies the second electric signal to a first floating diffusion layer according to a second control signal, a detection unit that detects whether or not a change amount of the first electric signal supplied to the connection node exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection, and a first pixel signal generation unit that generates a first pixel signal corresponding to the second electric signal supplied to the first floating diffusion layer, and a control method thereof. This brings about an effect that the detection result of detecting whether or not the change amount of the electric signal supplied to the connection node exceeds the threshold is output.

Furthermore, in the fifth aspect, a second photoelectric conversion element that photoelectrically converts incident light to generate a third electric signal, a third transistor that supplies to a second floating diffusion layer according to a third control signal, and a second pixel signal generation unit that generates a voltage signal corresponding to the third electric signal supplied to the second floating diffusion layer as a second pixel signal may be further provided. This brings about an effect that the pixel signals are sequentially generated in a case where the change amount exceeds the threshold.

Furthermore, a sixth aspect of the present technology is an imaging device provided with a first photoelectric conversion element that photoelectrically converts incident light to generate a first electric signal, a second photoelectric conversion element that photoelectrically converts the incident light to generate a second electric signal, a first signal supply unit that supplies the first electric signal to a connection node according to a first control signal, a second signal supply unit that supplies the second electric signal to the connection node according to a second control signal, a detection unit that detects whether or not a change amount of the electric signal supplied to the connection node exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection, and a recording unit that records the detection signal. This brings about an effect that the detection result of detecting whether or not the change amount of the electric signals from the plurality of photoelectric conversion elements exceeds the threshold is recorded.

Furthermore, a seventh aspect of the present technology is a solid-state imaging element provided with a first photoelectric conversion element that photoelectrically converts incident light to generate first and second electric signals, a second photoelectric conversion element that photoelectrically converts the incident light to generate third and fourth electric signals, a first detection unit that detects whether or not a change amount of the first electric signal exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection, a second detection unit that detects whether or not a change amount of the third electric signal exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection, a first transistor that supplies the first electric signal to the first detection unit according to a first control signal, a second transistor that supplies the third electric signal to the second detection unit according to a second control signal, a pixel signal generation unit that generates a pixel signal corresponding to any one of the second or fourth pixel signal, a third transistor that supplies the second electric signal to the pixel signal generation unit according to a third control signal, and a fourth transistor that supplies the fourth electric signal to the pixel signal generation unit according to a fourth control signal. This brings about an effect that the detection result of detecting whether or not the change amount of the electric signals from the plurality of photoelectric conversion elements exceeds the threshold is generated and the pixel signal is generated.

Effects of the Invention

According to the present technology, in a solid-state imaging element that detects an address event, an excellent effect that a circuit scale may be reduced may be obtained. Note that, the effects are not necessarily limited to the effects herein described and may be the effects described in the present disclosure.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a block diagram illustrating a configuration example of an imaging device in a first embodiment of the present technology.

FIG. **2** is a view illustrating an example of a stacking structure of a solid-state imaging element in the first embodiment of the present technology.

FIG. **3** is a block diagram illustrating a configuration example of the solid-state imaging element in the first embodiment of the present technology.

FIG. **4** is a block diagram illustrating a configuration example of a pixel array unit in the first embodiment of the present technology.

FIG. **5** is a circuit diagram illustrating a configuration example of a pixel block in the first embodiment of the present technology.

FIG. **6** is a block diagram illustrating a configuration example of an address event detection unit in the first embodiment of the present technology.

FIG. **7** is a circuit diagram illustrating a configuration example of a current/voltage conversion unit in the first embodiment of the present technology.

FIG. **8** is a circuit diagram illustrating a configuration example of a subtractor and a quantizer in the first embodiment of the present technology.

FIG. **9** is a block diagram illustrating a configuration example of a column analog-to-digital converter (ADC) in the first embodiment of the present technology.

FIG. **10** is a timing chart illustrating an example of an operation of the solid-state imaging element in the first embodiment of the present technology.

FIG. **11** is a flowchart illustrating an example of an operation of the solid-state imaging element in the first embodiment of the present technology.

FIG. **12** is a circuit diagram illustrating a configuration example of a pixel block in a first variation of the first embodiment of the present technology.

FIG. **13** is a circuit diagram illustrating a configuration example of a pixel block in a second variation of the first embodiment of the present technology.

FIG. **14** is a circuit diagram illustrating a configuration example of a pixel block in a third variation of the first embodiment of the present technology.

FIG. **15** is a block diagram illustrating a configuration example of a pixel array unit in a second embodiment of the present technology.

FIG. **16** is a circuit diagram illustrating a configuration example of a light reception unit in the second embodiment of the present technology.

FIG. **17** is a circuit diagram illustrating a configuration example of a light reception unit in which a transfer transistor is reduced in the second embodiment of the present technology.

FIG. **18** is a circuit diagram illustrating a configuration example of a current/voltage conversion unit in the second embodiment of the present technology.

FIG. **19** is a timing chart illustrating an example of an operation of a solid-state imaging element in the second embodiment of the present technology.

FIG. **20** is a circuit diagram illustrating a configuration example of a current/voltage conversion unit in a variation of the second embodiment of the present technology.

FIG. **21** is a circuit diagram illustrating a configuration example of an ADC in the variation of the second embodiment of the present technology.

FIG. **22** is a block diagram illustrating a configuration example of a pixel array unit in a third embodiment of the present technology.

FIG. **23** is a circuit diagram illustrating a configuration example of a light reception unit in the third embodiment of the present technology.

FIG. **24** is a block diagram illustrating a configuration example of an address event detection unit in the third embodiment of the present technology.

FIG. **25** is a circuit diagram illustrating a configuration example of a light reception unit in a variation of the third embodiment of the present technology.

FIG. **26** is a block diagram illustrating a configuration example of a pixel array unit in a fourth embodiment of the present technology.

FIG. **27** is a block diagram illustrating a configuration example of a pixel array unit in a variation of the fourth embodiment of the present technology.

FIG. **28** is a circuit diagram illustrating a configuration example of a normal pixel in a variation of the fourth embodiment of the present technology.

FIG. **29** is a block diagram illustrating a configuration example of a pixel array unit in a fifth embodiment of the present technology.

FIG. **30** is a block diagram illustrating a configuration example of a pixel block in the fifth embodiment of the present technology.

FIG. **31** is a block diagram illustrating a configuration example of a pixel block in a sixth embodiment of the present technology.

FIG. **32** is a block diagram illustrating a schematic configuration example of a vehicle control system.

FIG. **33** is an explanatory view illustrating an example of an installation position of an imaging unit.

MODE FOR CARRYING OUT THE INVENTION

Modes for carrying out the present technology (hereinafter, referred to as embodiments) are hereinafter described. The description is given in the following order.

1. First Embodiment (example in which a plurality of pixels shares an address event detection unit)
2. Second Embodiment (example in which pixel signal generation units are reduced and a plurality of pixels shares an address event detection unit)
3. Third Embodiment (example in which a plurality of pixels each provided with a capacitor shares an address event detection unit)
4. Fourth Embodiment (example in which an address event detection unit is arranged for each pixel)
5. Fifth Embodiment (example in which the number of pixels sharing an image signal generation unit is smaller than the number of pixels sharing an address event detection unit)
6. Application Example to Mobile Body

1. First Embodiment

[Configuration Example of Imaging Device]

FIG. **1** is a block diagram illustrating a configuration example of an imaging device **100** in a first embodiment of the present technology. The imaging device **100** is provided with an imaging lens **110**, a solid-state imaging element **200**, a recording unit **120**, and a control unit **130**. As the imaging device **100**, a camera mounted on an industrial robot, an in-vehicle camera and the like are assumed.

The imaging lens **110** condenses incident light and guides the same to the solid-state imaging element **200**. The solid-state imaging element **200** photoelectrically converts the incident light to image image data. The solid-state imaging element **200** executes predetermined signal processing such as image recognition processing on the imaged image data, and outputs data indicating a processing result and a detection signal of an address event to the recording unit **120** via a signal line **209**. A generating method of the detection signal is described later.

The recording unit **120** records the data from the solid-state imaging element **200**. The control unit **130** controls the solid-state imaging element **200** to image the image data.

[Configuration Example of Solid-State Imaging Element]

FIG. **2** is a view illustrating an example of a stacking structure of the solid-state imaging element **200** in the first embodiment of the present technology. The solid-state imaging element **200** is provided with a detection chip **202** and a light reception chip **201** stacked on the detection chip **202**. These chips are electrically connected to each other via a connection unit such as a via. Note that, they may also be connected to each other by Cu—Cu joint or a bump in addition to the via.

FIG. **3** is a block diagram illustrating a configuration example of the solid-state imaging element **200** in the first embodiment of the present technology. The solid-state imaging element **200** is provided with a drive circuit **211**, a signal processing unit **212**, an arbiter **213**, a column ADC **220**, and a pixel array unit **300**.

In the pixel array unit **300**, a plurality of pixels is arranged in a two-dimensional lattice manner. Furthermore, the pixel array unit **300** is divided into a plurality of pixel blocks each of which includes a predetermined number of pixels. Hereinafter, a set of pixels or pixel blocks arranged in a horizontal direction is referred to as a "row", and a set of pixels or pixel blocks arranged in a direction perpendicular to the row is referred to as a "column".

Each of the pixels generates an analog signal of a voltage corresponding to a photocurrent as a pixel signal. Furthermore, each of the pixel blocks detects presence/absence of the address event depending on whether or not a change amount of the photocurrent exceeds a predetermined threshold. Then, when the address event occurs, the pixel block outputs a request to the arbiter.

The drive circuit **211** drives each of the pixels to allow the same to output the pixel signal to the column ADC **220**.

The arbiter **213** arbitrates the request from each pixel block and transmits a response to the pixel block on the basis of an arbitration result. The pixel block that receives the response supplies a detection signal indicating a detection result to the drive circuit **211** and the signal processing unit **212**.

The column ADC **220** converts, for each column of the pixel blocks, the analog pixel signals from the column into digital signals. The column ADC **220** supplies the digital signals to the signal processing unit **212**.

The signal processing unit **212** executes predetermined signal processing such as correlated double sampling (CDS) processing and image recognition processing on the digital signals from the column ADC **220**. The signal processing unit **212** supplies data indicating a processing result and the detection signal to the recording unit **120** via the signal line **209**.

[Configuration Example of Pixel Array Unit]

FIG. **4** is a block diagram illustrating a configuration example of the pixel array unit **300** in the first embodiment of the present technology. The pixel array unit **300** is divided into a plurality of pixel blocks **310**. In each of the pixel blocks **310**, a plurality of pixels is arranged in I rows×J columns (I and J are integers).

Furthermore, the pixel block **310** is provided with a pixel signal generation unit **320**, a plurality of light reception units **330** of I rows×J columns, and an address event detection unit **400**. The plurality of light reception units **330** in the pixel block **310** shares the pixel signal generation unit **320** and the address event detection unit **400**. Then, a circuit including the light reception unit **330** at certain coordinates, the pixel signal generation unit **320**, and the address event detection unit **400** serves as a pixel at the coordinates. Furthermore, a vertical signal line VSL is arranged for each column of the pixel blocks **310**. When the number of columns of the pixel blocks **310** is set to m (m is an integer), m vertical signal lines VSL are arranged.

The light reception unit **330** photoelectrically converts the incident light to generate the photocurrent. The light reception unit **330** supplies the photocurrent to either the pixel signal generation unit **320** or the address event detection unit **400** under the control of the drive circuit **211**.

The pixel signal generation unit **320** generates a signal of the voltage corresponding to the photocurrent as a pixel signal SIG. The pixel signal generation unit **320** supplies the generated pixel signal SIG to the column ADC **220** via the vertical signal line VSL.

The address event detection unit **400** detects the presence/absence of the address event on the basis of whether or not the change amount of the photocurrent from each of the light reception units **330** exceeds a predetermined threshold. The address event includes, for example, an on-event indicating that the change amount exceeds an upper limit threshold and an off-event indicating that the change amount falls below a lower limit threshold. Furthermore, the detection signal of the address event includes, for example, one bit indicating a detection result of the on-event and one bit indicating a detection result of the off-event. Note that, it is also possible that the address event detection unit **400** detects only the on-event.

When the address event occurs, the address event detection unit **400** supplies a request for transmission of the detection signal to the arbiter **213**. Then, upon receiving a response to the request from the arbiter **213**, the address event detection unit **400** supplies the detection signal to the drive circuit **211** and the signal processing unit **212**. Note that, the address event detection unit **400** is an example of a detection unit recited in claims.

[Configuration Example of Pixel Block]

FIG. **5** is a circuit diagram illustrating a configuration example of the pixel block **310** in the first embodiment of the present technology. In the pixel block **310**, the pixel signal generation unit **320** is provided with a reset transistor **321**, an amplification transistor **322**, a selection transistor **323**, and a floating diffusion layer **324**. The plurality of light reception units **330** is commonly connected to the address event detection unit **400** via a connection node **340**.

Furthermore, each of the light reception units **330** is provided with a transfer transistor **331**, an overflow gate (OFG) transistor **332**, and a photoelectric conversion element **333**. When the number of pixels in the pixel block **310** is set to N (N is an integer), N transfer transistors **331**, N OFG transistors **332**, and N photoelectric conversion elements **333** are arranged. A transfer signal TRGn is supplied to an n-th (n is an integer from **1** to N) transfer transistor **331** in the pixel block **310** by the drive circuit **211**. A control signal OFGn is supplied to an n-th OFG transistor **332** by the drive circuit **211**.

Furthermore, as the reset transistor **321**, the amplification transistor **322**, and the selection transistor **323**, for example, N-type metal-oxide-semiconductor (MOS) transistors are used. The N-type MOS transistors are similarly used for the transfer transistor **331** and the OFG transistor **332**.

Furthermore, each of the photoelectric conversion elements **333** is arranged on the light reception chip **201**. All the elements other than the photoelectric conversion element **333** are arranged on the detection chip **202**.

The photoelectric conversion element **333** photoelectrically converts the incident light to generate charge. The transfer transistor **331** transfers the charge from the corresponding photoelectric conversion element **333** to the floating diffusion layer **324** according to the transfer signal TRGn. The OFG transistor **332** supplies an electric signal generated by the corresponding photoelectric conversion element **333** to the connection node **340** according to the control signal OFGn. Here, the supplied electric signal is the photocurrent including the charge. Note that, a circuit including the transfer transistor **331** and the OFG transistor **332** of each pixel is an example of a signal supply unit recited in claims.

The floating diffusion layer **324** accumulates the charge and generates a voltage corresponding to an amount of the accumulated charge. The reset transistor **321** initializes the charge amount of the floating diffusion layer **324** according to a reset signal from the drive circuit **211**. The amplification transistor **322** amplifies the voltage of the floating diffusion layer **324**. The selection transistor **323** outputs a signal of the amplified voltage as the pixel signal SIG to the column ADC **220** via the vertical signal line VSL according to a selection signal SEL from the drive circuit **211**.

When being instructed by the control unit **130** to start detecting the address event, the drive circuit **211** drives the OFG transistors **332** of all the pixels by the control signal OFGn to allow the same to supply the photocurrent. Therefore, the address event detection unit **400** is supplied with a current that is the sum of the photocurrents of all the light reception units **330** in the pixel block **310**.

Then, when the address event is detected in a certain pixel block **310**, the drive circuit **211** turns off all the OFG transistors **332** in this block to stop supplying the photocurrent to the address event detection unit **400**. Next, the drive circuit **211** sequentially drives the transfer transistors **331** by the transfer signal TRGn to transfer the charge to the floating diffusion layer **324**. Therefore, the pixel signals of the plurality of pixels in the pixel block **310** are sequentially output.

In this manner, the solid-state imaging element **200** outputs only the pixel signal of the pixel block **310** in which the address event is detected to the column ADC **220**. Therefore, it is possible to reduce power consumption of the solid-state imaging element **200** and a processing amount of the image processing as compared with a case where the pixel signals of all the pixels are output regardless of the presence/absence of the address event.

Furthermore, since a plurality of pixels shares the address event detection unit **400**, it is possible to reduce a circuit scale of the solid-state imaging element **200** as compared with a case where the address event detection unit **400** is arranged for each pixel.

[Configuration Example of Address Event Detection Unit]

FIG. **6** is a block diagram illustrating a configuration example of the address event detection unit **400** in the first embodiment of the present technology. The address event detection unit **400** is provided with a current/voltage conversion unit **410**, a buffer **420**, a subtractor **430**, a quantizer **440**, and a transfer unit **450**.

The current/voltage conversion unit **410** converts the photocurrent from the corresponding light reception unit **330** into a voltage signal of its logarithm. The current/voltage conversion unit **410** supplies the voltage signal to the buffer **420**.

The buffer **420** corrects the voltage signal from the current/voltage conversion unit **410**. The buffer **420** outputs the corrected voltage signal to the subtractor **430**.

The subtractor **430** lowers a level of the voltage signal from the buffer **420** according to a row drive signal from the drive circuit **211**. The subtractor **430** supplies the lowered voltage signal to the quantizer **440**.

The quantizer **440** quantizes the voltage signal from the subtractor **430** into a digital signal and outputs the same as the detection signal to the transfer unit **450**.

The transfer unit **450** transfers the detection signal from the quantizer **440** to the signal processing unit **212** and the like. When the address event is detected, the transfer unit **450** supplies the request for the transmission of the detection signal to the arbiter **213**. Then, upon receiving the response to the request from the arbiter **213**, the transfer unit **450** supplies the detection signal to the drive circuit **211** and the signal processing unit **212**.

[Configuration Example of Current/Voltage Conversion Unit]

FIG. **7** is a circuit diagram illustrating a configuration example of the current/voltage conversion unit **410** in the first embodiment of the present technology. The current/voltage conversion unit **410** is provided with N-type transistors **411** and **413** and a P-type transistor **412**. For example, MOS transistors are used as these transistors.

A source and a drain of the N-type transistor **411** are connected to the light reception unit **330** and a power supply terminal, respectively. The P-type transistor **412** and the N-type transistor **413** are connected in series between a power supply terminal and a ground terminal. Furthermore, a connection point between the P-type transistor **412** and the N-type transistor **413** is connected to a gate of the N-type transistor **411** and an input terminal of the buffer **420**. Furthermore, a predetermined bias voltage Vbias is applied to a gate of the P-type transistor **412**.

Drains of the N-type transistors **411** and **413** are connected to a power supply side, and such circuits are referred to as source followers. The photocurrent from the light reception unit **330** is converted into the voltage signal of its logarithm by the two source followers connected into a loop. Furthermore, the P-type transistor **412** supplies a constant current to the N-type transistor **413**.

[Configuration Example of Subtractor and Quantizer]

FIG. **8** is a circuit diagram illustrating a configuration example of the subtractor **430** and the quantizer **440** in the first embodiment of the present technology. The subtractor **430** is provided with capacitors **431** and **433**, an inverter **432**, and a switch **434**. Furthermore, the quantizer **440** is provided with a comparator **441**.

One end of the capacitor **431** is connected to an output terminal of the buffer **420** and the other end thereof is connected to an input terminal of the inverter **432**. The capacitor **433** is connected in parallel with the inverter **432**. The switch **434** opens/closes a path connecting both ends of the capacitor **433** according to the row drive signal.

The inverter **432** inverts the voltage signal input via the capacitor **431**. The inverter **432** outputs the inverted signal to a non-inverting input terminal (+) of the comparator **441**.

When the switch **434** is turned on, a voltage signal V_{init} is input to a buffer **420** side of the capacitor **431**, and the opposite side becomes a virtual ground terminal. Potential of this virtual ground terminal is set to zero for convenience. At that time, potential Q_{init} accumulated in the capacitor **431** is expressed by a following expression when capacitance of the capacitor **431** is set to C1. On the other hand, since both the ends of the capacitor **433** are short-circuited, the accumulated charge thereof is zero.

$$Q_{init} = C1 \times V_{init} \qquad \text{Expression 1}$$

Next, considering a case where the switch **434** is turned off and the voltage on the buffer **420** side of the capacitor **431** changes to V_{after}, charge Q_{after} accumulated in the capacitor **431** is expressed by a following expression.

$$Q_{after} = C1 \times V_{after} \qquad \text{Expression 2}$$

On the other hand, charge Q2 accumulated in the capacitor **433** is expressed by a following expression when an output voltage is set to V_{out}.

$$Q2 = -C2 \times V_{out} \quad \text{Expression 3}$$

At that time, a total charge amount of the capacitors **431** and **433** does not change, so that following expression holds.

$$Q_{init} = Q_{after} + Q2 \quad \text{Expression 4}$$

By substituting Expressions 1 to 3 into Expression 4 and transforming, a following expression is obtained.

$$V_{out} = -(C1/C2) \times (V_{after} - V_{init}) \quad \text{Expression 5}$$

Expression 5 expresses a subtracting operation of the voltage signal, and a gain of a subtraction result is **C1**/**C2**. Since it is generally desired to maximize the gain, it is preferable to design C**1** larger and C**2** smaller. On the other hand, if C**2** is too small, kTC noise increases, and there is a possibility that a noise characteristic deteriorates, so that a reduction in capacitance of C**2** is limited to a range in which the noise may be allowed. Furthermore, since the address event detection unit **400** including the subtractor **430** is mounted for each pixel block, there is a limitation in area of the capacitance C**1** and the capacitance C**2**. In consideration of them, values of the capacitance C**1** and the capacitance C**2** are determined.

The comparator **441** compares the voltage signal from the subtractor **430** with a predetermined threshold voltage Vth applied to an inverting input terminal (–). The comparator **441** outputs a signal indicating a comparison result to the transfer unit **450** as the detection signal.

Furthermore, a gain A of an entire address event detection unit **400** described above is expressed by a following expression when a conversion gain of the current/voltage conversion unit **410** is set to CG_{log} and a gain of the buffer **420** is set to "1".

[Mathematical Expression 1]

$$A == \frac{CG_{log} \cdot C1}{C2} \sum_{n=1}^{N} i_{photo}\_n \quad \text{Expression 6}$$

In the above expression, i_{photo}_n represents the photocurrent of the n-th pixel, and its unit is, for example, ampere (A). N represents the number of pixels in the pixel block **310**.

[Configuration Example of Column ADC]

FIG. **9** is a block diagram illustrating a configuration example of the column ADC **220** in the first embodiment of the present technology. The column ADC **220** is provided with an ADC **230** for each column of the pixel block **310**. Furthermore, the column ADC **220** is provided with a reference signal generation unit **231** and an output unit **232**. The reference signal generation unit **231** generates a reference signal such as a ramp signal and supplies the same to each of the ADCs **230**. A digital to analog converter (DAC) and the like is used as the reference signal generation unit **231**. The output unit **232** supplies the digital signal from the ADC **230** to the signal processing unit **212**.

The ADC **230** converts an analog pixel signal SIG supplied via the vertical signal line VSL into a digital signal. The ADC **230** is provided with a comparator **231**, a counter **232**, a switch **233**, and a memory **234**. The comparator **231** compares the reference signal with the pixel signal SIG, and the counter **232** counts a count value over a period until a comparison result is inverted. The switch **233** supplies the count value to the memory **234** and allows the same to hold the count value under the control of a timing control circuit (not illustrated) and the like. The memory **234** supplies a digital signal indicating the count value to the output unit **232** under the control of a horizontal drive unit (not illustrated) and the like. With this configuration, the pixel signal SIG is converted into a digital signal having a larger bit depth than that of the detection signal. For example, if the detection signal is of two bits, the pixel signal is converted into the digital signal of three or more bits (16 bits and the like). Note that, the ADC **230** is an example of an analog/digital converter recited in claims.

[Operation Example of Solid-State Imaging Element]

FIG. **10** is a timing chart illustrating an example of an operation of the solid-state imaging element **200** in the first embodiment of the present technology. At timing T**0**, when being instructed by the control unit **130** to start detecting the address event, the drive circuit **211** sets all the control signals OFGn to a high level to turn on the OFG transistors **332** of all the pixels. Therefore, the sum of the photocurrents of all the pixels is supplied to the address event detection unit **400**. On the other hand, all the transfer signals TRGn are at a low level, and the transfer transistors **331** of all the pixels are in an off state.

Then, suppose that the address event detection unit **400** detects the address event and outputs a high-level detection signal at timing T**1**. Here, suppose that the detection signal is a one-bit signal indicating the detection result of the on-event.

Upon receiving the detection signal, the drive circuit **211** sets all the control signals OFGn to a low level to stop supplying the photocurrent to the address event detection unit **400** at timing T**2**. Furthermore, the drive circuit **211** sets the selection signal SEL to a high level and sets a reset signal RST to a high level over a certain pulse period to initialize the floating diffusion layer **324**. The pixel signal generation unit **320** outputs a voltage at the initialization as a reset level, and the ADC **230** converts the reset level into a digital signal.

At timing T**3** after the conversion of the reset level, the drive circuit **211** supplies a high-level transfer signal TRG**1** over a certain pulse period to allow a first pixel to output a voltage as a signal level. The ADC **230** converts the signal level into a digital signal. The signal processing unit **212** obtains a difference between the reset level and the signal level as a net pixel signal. This processing is referred to as the CDS processing.

At timing T**4** after the conversion of the signal level, the drive circuit **211** supplies a high-level transfer signal TRG**2** over a certain pulse period to allow a second pixel to output a signal level. The signal processing unit **212** obtains a difference between the reset level and the signal level as a net pixel signal. Thereafter, similar processing is executed, and the pixel signals of the respective pixels in the pixel block **310** are sequentially output.

When all the pixel signals are output, the drive circuit **211** sets all the control signals OFGn to the high level and turns on the OFG transistors **332** of all the pixels.

FIG. **11** is a flowchart illustrating an example of the operation of the solid-state imaging element **200** in the first embodiment of the present technology. This operation starts, for example, when a predetermined application for detecting the address event is executed.

Each of the pixel blocks **310** detects the presence/absence of the address event (step S**901**). The drive circuit **211** determines whether or not there is the address event in any of the pixel blocks **310** (step S**902**). In a case where there is the address event (step S**902**: Yes), the drive circuit **211** allows the respective pixels in the pixel block **310** in which the address event occurs to sequentially output the pixel signals (step S**903**).

In a case where there is no address event (step S**902**: No) or after step S**903**, the solid-state imaging element **200** repeats step S**901** and subsequent steps.

In this manner, according to the first embodiment of the present technology, the address event detection unit **400** detects the change amount of the photocurrent of each of the plurality of (N) photoelectric conversion elements **333** (pixels), so that it is possible to arrange one address event detection unit **400** for every N pixels. By sharing one address event detection unit **400** by N pixels in this manner, the circuit scale may be reduced as compared with the configuration in which the address event detection unit **400** is not shared but provided for each pixel.

[First Variation]

In the above-described first embodiment, the elements other than the photoelectric conversion element **333** are arranged on the detection chip **202**, but in this configuration, there is a possibility that the circuit scale of the detection chip **202** increases as the number of pixels increases. A solid-state imaging element **200** in a first variation of the first embodiment is different from that in the first embodiment in that a circuit scale of a detection chip **202** is reduced.

FIG. **12** is a circuit diagram illustrating a configuration example of a pixel block **310** in the first variation of the first embodiment of the present technology. The pixel block **310** in the first variation of the first embodiment is different from that in the first embodiment in that a reset transistor **321**, a floating diffusion layer **324**, and a plurality of light reception units **330** are arranged on the light reception chip **201**. Other elements are arranged on a detection chip **202**.

In this manner, according to the first variation of the first embodiment of the present technology, since the reset transistor **321** and the like and a plurality of light reception units **330** are arranged on the light reception chip **201**, the circuit scale of the detection chip **202** may be reduced as compared with that in the first embodiment.

[Second Variation]

In the first variation of the first embodiment described above, the reset transistor **321** and the like and the plurality of light reception units **330** are arranged on the light reception chip **201**, but there is a possibility that the circuit scale of the detection chip **202** increases as the number of pixels increases. A solid-state imaging element **200** in a second variation of the first embodiment is different from that in the first variation of the first embodiment in that a circuit scale of a detection chip **202** is further reduced.

FIG. **13** is a circuit diagram illustrating a configuration example of a pixel block **310** in the second variation of the first embodiment of the present technology. The pixel block **310** in the second variation of the first embodiment is different from that in the first variation of the first embodiment in that N-type transistors **411** and **413** are further arranged on a light reception chip **201**. In this manner, by using only the N-type transistors in the light reception chip **201**, the number of steps of forming the transistors may be reduced as compared with a case where the N-type transistor and a P-type transistor are mixed. Therefore, a manufacturing cost of the light reception chip **201** may be reduced.

In this manner, according to the second variation of the first embodiment of the present technology, since the N-type transistors **411** and **413** are further arranged on the light reception chip **201**, the circuit scale of the detection chip **202** may be reduced as compared with that in the first variation of the first embodiment.

[Third Variation]

In the second variation of the first embodiment described above, the N-type transistors **411** and **413** are further arranged on the light reception chip **201**, but there is a possibility that the circuit scale of the detection chip **202** increases as the number of pixels increases. A solid-state imaging element **200** in a third variation of the first embodiment is different from that in the second variation of the first embodiment in that a circuit scale of a detection chip **202** is further reduced.

FIG. **14** is a circuit diagram illustrating a configuration example of a pixel block **310** in the third variation of the first embodiment of the present technology. The pixel block **310** in the third variation of the first embodiment is different from that in the second variation of the first embodiment in that an amplification transistor **322** and a selection transistor **323** are further arranged on a light reception chip **201**. That is, an entire pixel signal generation unit **320** is arranged on the light reception chip **201**.

In this manner, according to the third variation of the first embodiment of the present technology, since the pixel signal generation unit **320** is arranged on the light reception chip **201**, the circuit scale of the detection chip **202** may be reduced as compared with that in the second variation of the first embodiment.

2. Second Embodiment

Although the pixel signal generation unit **320** is provided for each pixel block **310** in the above-described first embodiment, there is a possibility that the circuit scale of the solid-state imaging element **200** increases as the number of pixels increases. A solid-state imaging element **200** in a second embodiment is different from that in the first embodiment in that pixel signal generation units **320** are reduced.

FIG. **15** is a block diagram illustrating a configuration example of a pixel array unit **300** in the second embodiment of the present technology. The pixel array unit **300** is different from that in the first embodiment in that the pixel signal generation unit **320** is not provided.

Furthermore, an address event detection unit **400** in the second embodiment is different from that in the first embodiment in generating a pixel signal SIG and outputting the same via a vertical signal line VSL.

FIG. **16** is a circuit diagram illustrating a configuration example of a light reception unit **330** in the second embodiment of the present technology. The light reception unit **330** in the second embodiment is different from that in the first embodiment in not including an OFG transistor **332**.

Furthermore, a transfer transistor **331** in the second embodiment supplies a photocurrent from a photoelectric conversion element **333** to the address event detection unit **400** via a connection node **340**.

Note that, although the transfer transistor **331** is arranged on each of the light reception units **330**, as illustrated in FIG. **17**, it is also possible to adopt a configuration without the transistors provided. In this case, a drive circuit **211** does not need to supply a transfer signal TRGn to the light reception unit **330**.

FIG. **18** is a circuit diagram illustrating a configuration example of a current/voltage conversion unit **410** in the second embodiment of the present technology. The current/voltage conversion unit **410** in the second embodiment is different from that in the first embodiment in that a source of an N-type transistor **413** is connected to the vertical signal line VSL.

Furthermore, when an address event is detected, the drive circuit **211** lowers a voltage (Vbias) to a gate of a P-type transistor **412** as compared with that before detection to a low level. Therefore, a voltage of a gate of an N-type transistor **411** reaches a power supply voltage VDD as that of a drain thereof, and the N-type transistor **411** is put into a state equivalent to a case of being diode-connected. Then, a pixel signal SIG of a voltage corresponding to the photocurrent is generated by the N-type transistor **413** that serves as a source follower.

Furthermore, a plurality of light reception units **330** and the N-type transistors **411** and **413** are arranged on a light reception chip **201**, and remaining elements are arranged on a detection chip **202**.

FIG. **19** is a timing chart illustrating an example of an operation of the solid-state imaging element **200** in the second embodiment of the present technology.

At timing T**0**, when being instructed to start detecting the address event, the drive circuit **211** sets all transfer signals TRGn to a high level to turn on the transfer transistors **331** of all the pixels.

Then, suppose that the address event detection unit **400** detects the address event and outputs a high-level detection signal at timing T**1**.

Upon receiving the detection signal, the drive circuit **211** sets only a transfer signal TRG**1** to a high level over a certain pulse period at timing T**2**. The pixel signal generation unit **320** converts a pixel signal of a first pixel into a digital signal.

At timing T**3** after the conversion of the pixel signal, the drive circuit **211** sets a high-level transfer signal TRG**2** to a high level over a certain pulse period. The pixel signal generation unit **320** converts a pixel signal of a second pixel into a digital signal. Thereafter, similar processing is executed, and the pixel signals of the respective pixels in the pixel block **310** are sequentially output.

When all the pixel signals are output, the drive circuit **211** sets all the transfer signals TRGn to the high level, and turns on the transfer transistors **331** of all the pixels.

In this manner, in the second embodiment of the present technology, the address event detection unit **400** generates the pixel signal SIG, so that it is not necessary to arrange the pixel signal generation unit **320**. Therefore, a circuit scale may be reduced as compared with that in the first embodiment in which the pixel signal generation unit **320** is arranged.

[Variation]

In the above-described second embodiment, an entire ADC **230** is arranged on the detection chip **202**; however, there is a possibility that the circuit scale of the detection chip **202** increases as the number of pixels increases. A solid-state imaging element **200** in a variation of the second embodiment is different from that in the second embodiment in that a part of an ADC **230** is arranged on a light reception chip **201** to reduce a circuit scale of a detection chip **202**.

FIG. **20** is a circuit diagram illustrating a configuration example of a current/voltage conversion unit **410** in the variation of the second embodiment of the present technology. The current/voltage conversion unit **410** in the variation of the second embodiment is different from that in the second embodiment in that a source of an N-type transistor **413** is grounded and a drain of an N-type transistor **411** is connected to a vertical signal line VSL. Note that, as in the second embodiment, it is also possible to connect the source of the N-type transistor **413** to the vertical signal line VSL in place of the N-type transistor **411**.

FIG. **21** is a circuit diagram illustrating a configuration example of the ADC **230** in the variation of the second embodiment of the present technology. The ADC **230** is provided with a differential amplification circuit **240** and a counter **250**.

The differential amplification circuit **240** is provided with N-type transistors **243**, **244**, and **245**, and P-type transistors **241** and **242**. For example, MOS transistors are used as these transistors.

The N-type transistors **243** and **244** form a differential pair, and sources of these transistors are commonly connected to a drain of the N-type transistor **245**. Furthermore, a drain of the N-type transistor **243** is connected to a drain of the P-type transistor **241** and gates of the P-type transistors **241** and **242**. A drain of the N-type transistor **244** is connected to a drain of the P-type transistor **242** and the counter **250**. Furthermore, a reference signal REF is input to a gate of the N-type transistor **243**, and a pixel signal SIG is input to a gate of the N-type transistor **244** via the vertical signal line VSL. Note that, the N-type transistor **243** is an example of a reference side transistor recited in claims, and the N-type transistor **244** is an example of a signal side transistor recited in claims.

For example, a ramp signal is used as the reference signal REF. A circuit that generates the reference signal REF is not illustrated.

A predetermined bias voltage Vb is applied to a gate of the N-type transistor **245** and a source thereof is grounded. This N-type transistor **245** supplies a constant current. Note that, the N-type transistor **245** is an example of a constant current source recited in claims.

With the above-described configuration, the P-type transistors **241** and **242** form a current mirror circuit, amplify a difference between the reference signal REF and the pixel signal SIG, and output the same to the counter **250**. Then, the counter **250** counts a count value over a period until a signal from the differential amplification circuit **240** is inverted, and outputs a digital signal indicating the count value to the signal processing unit **212**.

Furthermore, in the variation of the second embodiment described above, the light reception chip **201** is further provided with the above-described N-type transistors **243**, **244**, and **245**.

In this manner, according to the variation of the second embodiment of the present technology, since the N-type transistors **243**, **244**, and **245** are further arranged on the light reception chip **201**, the circuit scale of the detection chip **202** may be reduced as compared with that in the second embodiment.

3. Third Embodiment

In the above-described second embodiment, the capacitors **431** and **433** are arranged in the address event detection unit **400**; however, the gain is deteriorated when the capacitance C**1** is reduced according to expression 5, so that it is difficult to improve an operation speed of the circuit by reducing the capacitance C**1**. A solid-state imaging element

200 according to a third embodiment is different from that in the second embodiment in that a capacitor **431** is arranged for each pixel to improve an operation speed.

FIG. **22** is a block diagram illustrating a configuration example of a pixel array unit **300** in the third embodiment of the present technology. The pixel array unit **300** in the third embodiment is different from that in the second embodiment in that each of light reception units **330** generates a pixel signal SIG in place of an address event detection unit **400**. Furthermore, a vertical signal line VSL is wired for each column of pixels, for example. Then, an ADC **230** is also provided for each pixel column. Note that, as in the second embodiment, the vertical signal line VSL may be arranged for each column of pixel blocks **310** and each of the light reception units **330** may be connected thereto. In this case, the ADC **230** is also provided for each column of the pixel blocks **310**.

FIG. **23** is a circuit diagram illustrating a configuration example of the light reception unit **330** in the third embodiment of the present technology. The light reception unit **330** in the third embodiment is different from that in the second embodiment in further including a current/voltage conversion unit **410**, a buffer **420**, and a capacitor **431**.

A circuit configuration of the current/voltage conversion unit **410** in the third embodiment is similar to that in the variation of the second embodiment illustrated in FIG. **19**, for example. Furthermore, an operation of a drive circuit **211** in the third embodiment is similar to that in the second embodiment. Furthermore, circuits and elements arranged on a light reception chip **201** and a detection chip **202** in the third embodiment are similar to those in the variation of the second embodiment. That is, as illustrated in FIG. **20**, in the current/voltage conversion unit **410**, N-type transistors **411** and **413** are arranged on the light reception chip **201**. Furthermore, as illustrated in FIG. **21**, in the ADC **230**, N-type transistors **243**, **244**, and **245** are arranged on the light reception chip **201**.

FIG. **24** is a block diagram illustrating a configuration example of the address event detection unit **400** in the third embodiment of the present technology. The address event detection unit **400** in the third embodiment is different from that in the second embodiment in not including the current/voltage conversion unit **410**, the buffer **420**, and the capacitor **431**.

As described above, in the third embodiment, unlike the second embodiment in which a plurality of light reception units **330** connected in parallel shares one capacitor **431**, the capacitor **431** is provided for each light reception unit **330**. For this reason, individual capacitance of the capacitor **431** may be (C**1**)/N when the number of light reception units **330** (that is, the number of pixels) is set to N. By this reduction in capacitance, the operating speed of the circuit may be improved. However, an overall gain A in the third embodiment is expressed by a following expression.

[Mathematical Expression 2]

$$A == \frac{CG_{log} \cdot C1}{N \cdot C2} \sum_{n=1}^{N} i_{photo\_}n \qquad \text{Expression 7}$$

From Expressions 6 and 7, the gain A in the third embodiment is smaller than that in the first and second embodiments. For this reason, in exchange for improvement in operation speed, detection accuracy of the address event decreases.

In this manner, according to the third embodiment of the present technology, since the capacitor **431** is arranged for each of the light reception units **330**, it is possible to improve the operation speed of the circuit including the capacitor **431** as compared with a case where a plurality of light reception units **330** shares the capacitor **431**.

[Variation]

In the above-described third embodiment, the plurality of light reception units **330** (pixels) in the column shares one ADC **230**, but since it is required to sequentially convert the pixel signals of those pixels into digital signals, a read speed of the pixel signals decreases as the number of pixels in the column increases. A solid-state imaging element **200** in a variation of the third embodiment is different from that in the third embodiment in that an ADC **230** is arranged for each pixel.

FIG. **25** is a circuit diagram illustrating a configuration example of a light reception unit **330** in the variation of the third embodiment of the present technology. The light reception unit **330** in the variation of the third embodiment is different from that in the third embodiment in further including the ADC **230**.

In this manner, according to the variation of the third embodiment of the present technology, since the ADC **230** is arranged for each of the light reception units **330**, it is possible to improve a read speed of a pixel signal as compared with a configuration in which a plurality of light reception units **330** shares one ADC **230**.

4. Fourth Embodiment

In the above-described first embodiment, the address event is detected for each pixel block **310** including a plurality of pixels, but it is not possible to detect the address event occurring in each pixel. A solid-state imaging element **200** in a fourth embodiment is different from that in the first embodiment in that an address event detection unit **400** is arranged for each pixel.

FIG. **26** is a block diagram illustrating a configuration example of a pixel array unit **300** in the fourth embodiment of the present technology. The pixel array unit **300** in the fourth embodiment is different from that in the first embodiment in that a plurality of pixels **311** is arranged in a two-dimensional lattice manner. A pixel signal generation unit **320**, a light reception unit **330**, and the address event detection unit **400** are arranged in each of the pixels **311**. Circuit configurations of the pixel signal generation unit **320**, the light reception unit **330**, and the address event detection unit **400** are similar to those in the first embodiment.

Furthermore, circuits and elements arranged in the light reception chip **201** and the detection chip **202** are similar to those in any of the first embodiment and the first, second, and third variations of the first embodiment. For example, as illustrated in FIG. **5**, only a photoelectric conversion element **333** is arranged on a light reception chip **201**, and remaining elements are arranged on a detection chip **202**.

In this manner, according to the fourth embodiment of the present technology, since the address event detection unit **400** is arranged for each pixel, the address event may be detected for each pixel. Therefore, resolution of detection data of the address event may be improved as compared with a case where the address event is detected for each pixel block **310**.

[Variation]

Although the address event detection unit **400** is arranged in all the pixels in the above-described fourth embodiment, it is possible that the circuit scale of the solid-state imaging element **200** increases as the number of pixels increases. A solid-state imaging element **200** in a variation of the fourth embodiment is different from that in the fourth embodiment in that an address event detection unit **400** is arranged only in a pixel being a detection target out of a plurality of pixels.

FIG. **27** is a block diagram illustrating a configuration example of a pixel array unit **300** in the variation of the fourth embodiment of the present technology. The pixel array unit **300** in the variation of the fourth embodiment is different from that in the fourth embodiment in that a pixel in which the address event detection unit **400** is not arranged and a pixel in which the address event detection unit **400** is arranged are arranged. The former is a normal pixel **312** and the latter is an address event detection pixel **313**. The address event detection pixels **313** are arranged at regular intervals, for example. Note that, a plurality of address event detection pixels **313** may be arranged so as to be adjacent to each other.

Furthermore, a configuration of the address event detection pixel **313** is similar to that of the pixel **311** in the fourth embodiment. The normal pixel **312** is described later in detail.

FIG. **28** is a circuit diagram illustrating a configuration example of the normal pixel **312** in the variation of the fourth embodiment of the present technology. The normal pixel **312** in the variation of the fourth embodiment is provided with a photoelectric conversion element **333**, a transfer transistor **331**, a reset transistor **321**, an amplification transistor **322**, a selection transistor **323**, and a floating diffusion layer **324**. A connection configuration of these elements is similar to that in the first embodiment illustrated in FIG. **5**.

In this manner, according to the variation of the fourth embodiment of the present technology, since the address event detection unit **400** is arranged only in the address event detection pixel **313** among all the pixels, a circuit scale may be reduced as compared with the configuration in which the address event detection unit **400** is arranged in all the pixels.

5. Fifth Embodiment

In the first embodiment described above, the number of pixels sharing the address event detection unit **400** and the number of pixels sharing the image signal generation unit **320** are made the same, but the latter may be reduced. A solid-state imaging element **200** in a fifth embodiment is different from that in the first embodiment in that the number of pixels sharing an image signal generation unit **320** is smaller than the number of pixels sharing an address event detection unit **400**.

FIG. **29** is a block diagram illustrating a configuration example of a pixel array unit **300** in the fifth embodiment of the present technology. In the pixel array unit **300** in the fifth embodiment, N light reception units **330** (pixels) and one address event detection unit **400** are arranged in each of pixel blocks **310**. Furthermore, in each of the pixel blocks **310**, the pixel signal generation unit **320** is arranged for every M (M is an integer smaller than N) light reception units **330** (pixels).

FIG. **30** is a block diagram illustrating a configuration example of the pixel block **310** in the fifth embodiment of the present technology. In each of the pixel blocks **310**, the N light reception units **330** (pixels) share one address event detection unit **400**. Furthermore, the M pixels share one image signal generation unit **320**. The image signal generation unit **320** generates the pixel signal of a selected pixel out of the corresponding M pixels.

In this manner, according to the fifth embodiment of the present technology, the number of pixels sharing the image signal generation unit **320** is made smaller than the number of pixels sharing the address event detection unit **400**, so that it is possible to improve a read speed of a pixel signal as compared with a case where they made the same.

6. Sixth Embodiment

In the first embodiment described above, a plurality of pixels shares the image signal generation unit **320** and the address event detection unit **400**; however, it is also possible to arrange the address event detection unit **400** for each pixel. A solid-state imaging element **200** in a sixth embodiment is different from that in the first embodiment in that an address event detection unit **400** is arranged for each pixel while a plurality of pixels shares an image signal generation unit **320**.

FIG. **31** is a block diagram illustrating a configuration example of a pixel block **310** in the sixth embodiment of the present technology. In each of the pixel blocks **310**, the N light reception units **330** (pixels) share one pixel signal generation unit **320**. On the other hand, the address event detection unit **400** is arranged for each light reception unit **330** (pixel), and the light reception unit **330** is connected to a corresponding address event detection unit **400**.

In this manner, according to the sixth embodiment of the present technology, since the address event detection unit **400** is arranged for each pixel, presence/absence of an address event may be detected for each pixel.

<7. Application Example to Mobile Body>

The technology according to the present disclosure (present technology) is applicable to various products. For example, the technology according to the present disclosure may also be realized as a device mounted on any type of mobile body such as an automobile, an electric automobile, a hybrid electric automobile, a motorcycle, a bicycle, a personal mobility, an airplane, a drone, a ship, and a robot.

FIG. **32** is a block diagram illustrating a schematic configuration example of a vehicle control system that is an example of a mobile body control system to which the technology according to the present disclosure may be applied.

A vehicle control system **12000** is provided with a plurality of electronic control units connected to one another via a communication network **12001**. In the example illustrated in FIG. **32**, the vehicle control system **12000** is provided with a drive system control unit **12010**, a body system control unit **12020**, a vehicle exterior information detection unit **12030**, a vehicle interior information detection unit **12040**, and an integrated control unit **12050**. Furthermore, a microcomputer **12051**, an audio image output unit **12052**, and an in-vehicle network interface (I/F) **12053** are illustrated as functional configurations of the integrated control unit **12050**.

The drive system control unit **12010** controls operation of devices related to a drive system of a vehicle according to various programs. For example, the drive system control unit **12010** serves as a control device of a driving force generating device for generating driving force of the vehicle such as an internal combustion engine or a driving motor, a driving force transmitting mechanism for transmitting the driving force to wheels, a steering mechanism for adjusting a rudder angle of the vehicle, a braking device for generating braking force of the vehicle and the like.

The body system control unit **12020** controls operation of various devices mounted on a vehicle body in accordance with the various programs. For example, the body system control unit **12020** serves as a control device of a keyless entry system, a smart key system, a power window device, or various lights such as a head light, a backing light, a brake light, a blinker, or a fog light. In this case, a radio wave transmitted from a portable device that substitutes for a key or signals of various switches may be input to the body system control unit **12020**. The body system control unit **12020** receives an input of the radio wave or signals and controls a door lock device, a power window device, the lights and the like of the vehicle.

The vehicle exterior information detection unit **12030** detects information outside the vehicle on which the vehicle control system **12000** is mounted. For example, an imaging unit **12031** is connected to the vehicle exterior information detection unit **12030**. The vehicle exterior information detection unit **12030** allows the imaging unit **12031** to take an image of the exterior of the vehicle and receives the taken image. The vehicle exterior information detection unit **12030** may perform detection processing of objects such as a person, a vehicle, an obstacle, a sign, or a character on a road surface or distance detection processing on the basis of the received image.

The imaging unit **12031** is an optical sensor that receives light and outputs an electric signal corresponding to an amount of the received light. The imaging unit **12031** may output the electric signal as the image or output the same as ranging information. Furthermore, the light received by the imaging unit **12031** may be visible light or invisible light such as infrared light.

The vehicle interior information detection unit **12040** detects information in the vehicle. The vehicle interior information detection unit **12040** is connected to, for example, a driver state detection unit **12041** for detecting a state of a driver. The driver state detection unit **12041** includes, for example, a camera that images the driver, and the vehicle interior information detection unit **12040** may calculate a driver's fatigue level or concentration level or may determine whether or not the driver is dozing on the basis of detection information input from the driver state detection unit **12041**.

The microcomputer **12051** may calculate a control target value of the driving force generating device, the steering mechanism, or the braking device on the basis of the information inside and outside the vehicle obtained by the vehicle exterior information detection unit **12030** or the vehicle interior information detection unit **12040**, and output a control instruction to the drive system control unit **12010**. For example, the microcomputer **12051** may perform cooperative control for realizing functions of advanced driver assistance system (ADAS) including collision avoidance or impact attenuation of the vehicle, following travel based on the distance between the vehicles, vehicle speed maintaining travel, vehicle collision warning, vehicle lane departure warning and the like.

Furthermore, the microcomputer **12051** may perform the cooperative control for realizing automatic driving and the like to autonomously travel independent from the operation of the driver by controlling the driving force generating device, the steering mechanism, the braking device and the like on the basis of the information around the vehicle obtained by the vehicle exterior information detection unit **12030** or the vehicle interior information detection unit **12040**.

Furthermore, the microcomputer **12051** may output the control instruction to the body system control unit **12020** on the basis of the information outside the vehicle obtained by the vehicle exterior information detection unit **12030**. For example, the microcomputer **12051** may perform the cooperative control to realize glare protection such as controlling the head light according to a position of a preceding vehicle or an oncoming vehicle detected by the vehicle exterior information detection unit **12030** to switch a high beam to a low beam.

The audio image output unit **12052** transmits at least one of audio or image output signal to an output device capable of visually or audibly notifying an occupant of the vehicle or the outside the vehicle of the information. In the example in FIG. **32**, as the output device, an audio speaker **12061**, a display unit **12062**, and an instrument panel **12063** are illustrated. The display unit **12062** may include at least one of an on-board display or a head-up display, for example.

FIG. **33** is a view illustrating an example of an installation position of the imaging unit **12031**.

In FIG. **33**, imaging units **12101**, **12102**, **12103**, **12104**, and **12105** are included as the imaging unit **12031**.

The imaging units **12101**, **12102**, **12103**, **12104**, and **12105** are provided in positions such as, for example, a front nose, a side mirror, a rear bumper, a rear door, and an upper portion of a front windshield in a vehicle interior of the vehicle **12100**. The imaging unit **12101** provided on the front nose and the imaging unit **12105** provided in the upper portion of the front windshield in the vehicle interior principally obtain images in front of the vehicle **12100**. The imaging units **12102** and **12103** provided on the side mirrors principally obtain images of the sides of the vehicle **12100**. The imaging unit **12104** provided on the rear bumper or the rear door principally obtains an image behind the vehicle **12100**. The imaging unit **12105** provided on the upper portion of the front windshield in the vehicle interior is mainly used for detecting the preceding vehicle, a pedestrian, an obstacle, a traffic signal, a traffic sign, a lane and the like.

Note that, in FIG. **33**, an example of imaging ranges of the imaging units **12101** to **12104** is illustrated. An imaging range **12111** indicates the imaging range of the imaging unit **12101** provided on the front nose, imaging ranges **12112** and **12113** indicate the imaging ranges of the imaging units **12102** and **12103** provided on the side mirrors, and an imaging range **12114** indicates the imaging range of the imaging unit **12104** provided on the rear bumper or the rear door. For example, image data taken by the imaging units **12101** to **12104** are superimposed, so that an overlooking image of the vehicle **12100** as seen from above is obtained.

At least one of the imaging units **12101** to **12104** may have a function of obtaining distance information. For example, at least one of the imaging units **12101** to **12104** may be a stereo camera including a plurality of imaging elements, or may be an imaging element including pixels for phase difference detection.

For example, the microcomputer **12051** may extract especially a closest solid object on a traveling path of the vehicle **12100**, the solid object traveling at a predetermined speed (for example, 0 km/h or higher) in a direction substantially the same as that of the vehicle **12100** as the preceding vehicle by obtaining a distance to each solid object in the imaging ranges **12111** to **12114** and a change in time of the distance (relative speed relative to the vehicle **12100**) on the basis of the distance information obtained from the imaging units **12101** to **12104**. Moreover, the microcomputer **12051** may set the distance between the vehicles to be secured in advance from the preceding vehicle, and may perform automatic brake control (including following stop control), automatic acceleration control (including following start control) and the like. In this manner, it is possible to perform the cooperative control for realizing the automatic driving and the like to autonomously travel independent from the operation of the driver.

For example, the microcomputer **12051** may extract solid object data regarding the solid object while sorting the same into a motorcycle, a standard vehicle, a large-sized vehicle, a pedestrian, and other solid objects such as a utility pole on the basis of the distance information obtained from the imaging units **12101** to **12104** and use for automatically avoiding obstacles. For example, the microcomputer **12051** discriminates the obstacles around the vehicle **12100** into an obstacle visible to a driver of the vehicle **12100** and an obstacle difficult to see. Then, the microcomputer **12051** determines a collision risk indicating a degree of risk of collision with each obstacle, and when the collision risk is equal to or higher than a set value and there is a possibility of collision, this may perform driving assistance for avoiding the collision by outputting an alarm to the driver via the audio speaker **12061** and the display unit **12062** or performing forced deceleration or avoidance steering via the drive system control unit **12010**.

At least one of the imaging units **12101** to **12104** may be an infrared camera for detecting infrared rays. For example, the microcomputer **12051** may recognize a pedestrian by determining whether or not there is a pedestrian in the images taken by the imaging units **12101** to **12104**. Such pedestrian recognition is carried out, for example, by a procedure of extracting feature points in the images taken by the imaging units **12101** to **12104** as the infrared cameras and a procedure of performing pattern matching processing on a series of feature points indicating an outline of an object to discriminate whether or not this is a pedestrian. When the microcomputer **12051** determines that there is a pedestrian in the images taken by the imaging units **12101** to **12104** and recognizes the pedestrian, the audio image output unit **12052** controls the display unit **12062** to superimpose a rectangular contour for emphasis on the recognized pedestrian. Furthermore, the audio image output unit **12052** may control the display unit **12062** to display an icon and the like indicating the pedestrian at a desired position.

An example of the vehicle control system to which the technology according to the present disclosure may be applied is described above. The technology according to the present disclosure may be applied to the imaging unit **12031**, for example, out of the configurations described above. Specifically, the imaging device **100** in FIG. **1** may be applied to the imaging unit **12031**. By applying the technology according to the present disclosure to the imaging unit **12031**, it is possible to reduce the circuit mounting area and downsize the imaging unit **12031**.

Note that, the above-described embodiments describe an example of embodying the present technology, and there is a correspondence relationship between matters in the embodiments and the matters specifying the invention in claims. Similarly, there is a correspondence relationship between the matters specifying the invention in claims and the matters in the embodiments of the present technology assigned with the same names. However, the present technology is not limited to the embodiments and may be embodied with various modifications of the embodiment without departing from the spirit thereof.

Furthermore, the procedures described in the above-described embodiments may be considered as a method including a series of procedures and may be considered as a program for allowing a computer to execute the series of procedures and a recording medium which stores the program. A compact disc (CD), a MiniDisc (MD), a digital versatile disc (DVD), a memory card, a Blu-Ray™ Disc and the like may be used, for example, as the recording medium.

Note that, the effect described in this specification is illustrative only and is not limitative; there may also be another effect.

Note that, the present technology may also have a following configuration.

(1) A solid-state imaging element provided with:
- a plurality of photoelectric conversion elements each of which photoelectrically converts incident light to generate a first electric signal; and
- a detection unit that detects whether or not a change amount of the first electric signal of each of the plurality of photoelectric conversion elements exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection.

(2) The solid-state imaging element according to (1) described above, further provided with:
- a signal supply unit that supplies the first electric signal of each of the plurality of photoelectric conversion elements to a connection node according to a predetermined control signal,
- in which the detection unit detects whether or not the change amount of the first electric signal supplied to the connection node exceeds the predetermined threshold.

(3) The solid-state imaging element according to (2) described above, further provided with:
- a pixel signal generation unit that generates a pixel signal according to a second electric signal generated by the photoelectric conversion element,
- in which the signal supply unit sequentially selects the second electric signal of each of the plurality of photoelectric conversion elements to supply to the pixel signal generation unit in a case where the change amount exceeds the predetermined threshold.

(4) The solid-state imaging element according to (3) described above,
- in which the connection node is connected to N (N is an integer not smaller than 2) of the photoelectric conversion elements, and
- the pixel signal generation unit generates a signal of a voltage corresponding to the second electric signal of an element selected according to a selection signal out of M (M is an integer smaller than N) of the photoelectric conversion elements as the pixel signal.

(5) The solid-state imaging element according to (3) described above,
- in which the pixel signal generation unit is provided with:
- a reset transistor that initializes a floating diffusion layer;
- an amplification transistor that amplifies a signal of a voltage of the floating diffusion layer; and
- a selection transistor that outputs the amplified signal as the pixel signal according to a selection signal, and
- the detection unit is provided with:
- a plurality of N-type transistors that converts the first electric signal into a voltage signal of a logarithm of the first electric signal; and
- a P-type transistor that supplies a constant current to the plurality of N-type transistors.

(6) The solid-state imaging element according to (5) described above,
- in which the plurality of photoelectric conversion elements is arranged on a light reception chip, and

the detection unit and the pixel signal generation unit are arranged on a detection chip stacked on the light reception chip.

(7) The solid-state imaging element according to (5) described above,

in which the plurality of photoelectric conversion elements and the reset transistor are arranged on a light reception chip, and

the detection unit, the amplification transistor, and the selection transistor are arranged on a detection chip stacked on the light reception chip.

(8) The solid-state imaging element according to (5) described above,

in which the plurality of photoelectric conversion elements, the reset transistor, and the plurality of N-type transistors are arranged on a light reception chip, and

the amplification transistor, the selection transistor, and the P-type transistor are arranged on a detection chip stacked on the light reception chip.

(9) The solid-state imaging element according to (5) described above,

in which the plurality of photoelectric conversion elements, the pixel signal generation unit, and the plurality of N-type transistors are arranged on a light reception chip, and

the P-type transistor is arranged on a detection chip stacked on the light reception chip.

(10) The solid-state imaging element according to (1) described above, further provided with:

a signal supply unit that supplies the first electric signal of each of the plurality of photoelectric conversion elements to a connection node according to a predetermined control signal,

in which the detection unit further outputs a pixel signal corresponding to the first electric signal,

the signal supply unit sequentially selects the first electric signal of each of the plurality of photoelectric conversion elements to supply to the connection node in a case where the change amount exceeds the predetermined threshold, and

the detection unit is provided with:

first and second N-type transistors that convert the first electric signal into a voltage signal of a logarithm of the first electric signal; and

a P-type transistor that supplies a constant current to the first and second N-type transistors.

(11) The solid-state imaging element according to (10) described above, further provided with:

an analog/digital converter that converts the pixel signal into a digital signal,

in which the plurality of photoelectric conversion elements, the signal supply unit, and the first and second N-type transistors are arranged on a light reception chip, and

the P-type transistor and at least a part of the analog/digital converter are arranged on a detection chip stacked on the light reception chip.

(12) The solid-state imaging element according to (11) described above,

in which the analog/digital converter is provided with:

a signal side transistor to which the pixel signal is input;

a reference side transistor to which a predetermined reference signal is input;

a constant current source connected to the signal side transistor and the reference side transistor; and

a current mirror circuit that amplifies a difference between the pixel signal and the predetermined reference signal to output,

the plurality of photoelectric conversion elements, the signal supply unit, the first and second N-type transistors, the signal side transistor, the reference side transistor, and the constant current source are arranged on a light reception chip, and

the P-type transistor and the current mirror circuit are arranged on a detection chip stacked on the light reception chip.

(13) The solid-state imaging element according to (1) described above, further provided with:

a connection node connected to the photoelectric conversion element and the detection unit; and

for each of the plurality of photoelectric conversion elements, a current/voltage conversion unit that converts a photocurrent into a voltage signal of a logarithm of the photocurrent, a buffer that corrects the voltage signal to output, a capacitor inserted between the buffer and the connection node, and a signal processing unit that supplies an electric signal of each of the plurality of photoelectric conversion elements to the connection node through the current/voltage conversion unit, the buffer, and the capacitor according to a predetermined control signal,

in which the electric signal includes the photocurrent and the voltage signal.

(14) The solid-state imaging element according to (13) described above, further provided with:

an analog/digital converter that converts a pixel signal into a digital signal,

in which each of a predetermined number of current/voltage conversion units arranged in a predetermined direction further generates a signal of a voltage corresponding to the photocurrent as the pixel signal, and outputs the pixel signal to the analog/digital converter.

(15) The solid-state imaging element according to (13) described above, further provided with:

an analog/digital converter that converts a pixel signal into a digital signal for each of the plurality of photoelectric conversion elements,

in which each of current/voltage conversion units further generates a signal of a voltage corresponding to the photocurrent as the pixel signal, and outputs the pixel signal to the analog/digital converter.

(16) A solid-state imaging element provided with:

a photoelectric conversion element that photoelectrically converts incident light to generate an electric signal;

a signal supply unit that supplies the electric signal to either a connection node or a floating diffusion layer according to a predetermined control signal;

a detection unit that detects whether or not a change amount of the electric signal supplied to the connection node exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection; and

a pixel signal generation unit that generates a voltage signal corresponding to the electric signal supplied to the floating diffusion layer as a pixel signal.

(17) The solid-state imaging element according to (16) described above,

in which the signal supply unit includes:

a first transistor that supplies the electric signal to the connection node according to a predetermined control signal; and a second transistor that supplies the electric signal to a floating diffusion layer according to a predetermined control signal,

the pixel signal generation unit is arranged in each of a plurality of pixels, and

the first transistor and the detection unit are arranged in a pixel being a detection target out of the plurality of pixels.

(18) An imaging device provided with:

a plurality of photoelectric conversion elements each of which photoelectrically converts incident light to generate an electric signal;

a signal supply unit that supplies the electric signal of each of the plurality of photoelectric conversion elements to a connection node according to a predetermined control signal;

a detection unit that detects whether or not a change amount of the electric signal supplied to the connection node exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection; and

a recording unit that records the detection signal.

(19) A control method of a solid-state imaging element, the method provided with:

a signal supplying step of supplying, to a connection node, an electric signal of each of a plurality of photoelectric conversion elements each of which photoelectrically converts incident light to generate the electric signal according to a predetermined control signal; and

a detecting step of detecting whether or not a change amount of the electric signal supplied to the connection node exceeds a predetermined threshold and outputting a detection signal indicating a result of the detection.

(20) A solid-state imaging element provided with:

a first photoelectric conversion element that generates a first electric signal;

a second photoelectric conversion element that generates a second electric signal;

a detection unit that detects whether or not at least any one of a change amount of the first electric signal or a change amount of the second electric signal exceeds a predetermined threshold to output a detection signal indicating a result of the detection; and

a connection node connected to the first photoelectric conversion element, the second photoelectric conversion element, and the detection unit.

(21) The solid-state imaging element according to (20) described above, further provided with:

a first transistor that supplies the first electric signal to the connection node according to a first control signal; and

a second transistor that supplies the second electric signal to the connection node according to a second control signal,

in which the detection unit detects whether or not a change amount of either the first or second electric signal supplied to the connection node exceeds the predetermined threshold.

(22) The solid-state imaging element according to (21) described above, further provided with:

a pixel signal generating unit that generates a first pixel signal according to a third electric signal generated by the first photoelectric conversion element and generates a second pixel signal according to a fourth electric signal generated by the second photoelectric conversion element;

a third transistor connected to the first photoelectric conversion element and the pixel signal generating unit; and

a fourth transistor connected to the second photoelectric conversion element and the pixel signal generating unit,

in which the third transistor supplies the third electric signal to the pixel signal generating unit in a case where the change amount of the first electric signal exceeds the predetermined threshold, and

the fourth transistor supplies the fourth electric signal to the pixel signal generating unit in a case where the change amount of the second electric signal exceeds the predetermined threshold

(23) The solid-state imaging element according to (22) described above,

in which the pixel signal generating unit includes:

a first pixel signal generation unit that generates a first pixel signal according to the third electric signal generated by the first photoelectric conversion element; and

a second pixel signal generation unit that generates a second pixel signal according to the fourth electric signal generated by the second photoelectric conversion element,

the third transistor supplies the third electric signal to the first pixel signal generation unit in a case where the change amount of the first electric signal exceeds the predetermined threshold, and

the fourth transistor supplies the fourth electric signal to the second pixel signal generation unit in a case where the change amount of the second electric signal exceeds the predetermined threshold

(24) The solid-state imaging element according to (22) described above, further provided with:

a third photoelectric conversion element that generates a fifth electric signal and a sixth electric signal;

a fifth transistor that supplies an electric signal of the fifth photoelectric conversion element to the connection node according to a third control signal; and

a second pixel signal generating unit that generates a third pixel signal according to the sixth electric signal,

in which the sixth transistor supplies the sixth electric signal to the second pixel signal generating unit in a case where a change amount of the fifth electric signal exceeds the predetermined threshold.

(25) The solid-state imaging element according to (22) described above,

in which the pixel signal generating unit is provided with:

a reset transistor that initializes a floating diffusion layer;

an amplification transistor that amplifies a signal of a voltage of the floating diffusion layer; and

a selection transistor that outputs the amplified signal as the first or second pixel signal according to a selection signal, and

the detection unit is provided with:

a plurality of N-type transistors that converts a photocurrent into a voltage signal of a logarithm of the photocurrent; and

a P-type transistor that supplies a constant current to the plurality of N-type transistors.

(26) The solid-state imaging element according to (20) described above,

the first electric signal including a first photocurrent, and

the second electric signal including a second photocurrent,

the solid-state imaging element further provided with:

a connection node connected to the first photoelectric conversion element, the second photoelectric conversion element, and the detection unit;

a first current/voltage conversion unit that converts at least one of the first photocurrent or the second photocurrent into a voltage signal of a logarithm of the photocurrent;

a buffer that corrects the voltage signal to output;

a capacitor inserted between the buffer and the connection node; and

a signal processing unit that supplies at least one of the first electric signal or the second electric signal to the connection node through the current/voltage conversion unit, the buffer, and the capacitor according to a predetermined control signal,

in which the first photoelectric conversion element generates the first photocurrent, and

the second photoelectric conversion element generates the second photocurrent.

(27) The solid-state imaging element according to (26) described above, further provided with:

an analog/digital converter connected to the first current/voltage conversion unit and the second current/voltage conversion unit,

in which the first current/voltage conversion unit further generates a signal of a voltage corresponding to the first photocurrent as a first pixel signal, and outputs the first pixel signal to the analog/digital converter, and

the second current/voltage conversion unit further generates a signal of a voltage corresponding to the second photocurrent as a second pixel signal, and outputs the second pixel signal to the analog/digital converter.

(28) The solid-state imaging element according to (26) described above, further provided with:

a first analog/digital converter that converts a first pixel signal into a first digital signal; and

a second analog/digital converter that converts a second pixel signal into a second digital signal,

in which the first current/voltage conversion unit further generates a signal of a voltage corresponding to the first photocurrent as the first pixel signal, and outputs the first pixel signal to the first analog/digital converter, and

the second current/voltage conversion unit further generates a signal of a voltage corresponding to the second photocurrent as the second pixel signal, and outputs the second pixel signal to the second analog/digital converter.

(29) A solid-state imaging element provided with:

a first photoelectric conversion element that photoelectrically converts incident light to generate a first electric signal and a second electric signal;

a first signal supply unit that supplies the first electric signal to a connection node according to a first control signal;

a second signal supply unit that supplies the second electric signal to a first floating diffusion layer according to a second control signal;

a detection unit that detects whether or not a change amount of the first electric signal supplied to the connection node exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection; and

a first pixel signal generation unit that generates a first pixel signal corresponding to the second electric signal supplied to the first floating diffusion layer.

(30) The solid-state imaging element according to (29) described above, further provided with:

a second photoelectric conversion element that photoelectrically converts incident light to generate a third electric signal;

a third transistor that supplies to a second floating diffusion layer according to a third control signal; and

a second pixel signal generation unit that generates a voltage signal corresponding to the third electric signal supplied to the second floating diffusion layer as a second pixel signal.

(31) An imaging device provided with:

a first photoelectric conversion element that photoelectrically converts incident light to generate a first electric signal;

a second photoelectric conversion element that photoelectrically converts the incident light to generate a second electric signal;

a first signal supply unit that supplies the first electric signal to a connection node according to a first control signal;

a second signal supply unit that supplies the second electric signal to the connection node according to a second control signal;

a detection unit that detects whether or not a change amount of the electric signal supplied to the connection node exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection; and

a recording unit that records the detection signal.

(32) A control method of a solid-state imaging element, the method provided with:

a signal supplying step of supplying a first electric signal generated by photoelectric conversion of incident light by a first photoelectric conversion element to a connection node according to a first control signal and supplying a second electric signal generated by photoelectric conversion of the incident light by a second photoelectric conversion element to the connection node according to a second control signal; and

a detecting step of detecting whether or not a change amount of the electric signal supplied to the connection node exceeds a predetermined threshold and outputting a detection signal indicating a result of the detection.

(33) A solid-state imaging element provided with:

a first photoelectric conversion element that photoelectrically converts incident light to generate first and second electric signals;

a second photoelectric conversion element that photoelectrically converts the incident light to generate third and fourth electric signals;

a first detection unit that detects whether or not a change amount of the first electric signal exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection;

a second detection unit that detects whether or not a change amount of the third electric signal exceeds a predetermined threshold and outputs a detection signal indicating a result of the detection;

a first transistor that supplies the first electric signal to the first detection unit according to a first control signal;

a second transistor that supplies the third electric signal to the second detection unit according to a second control signal;

a pixel signal generation unit that generates a pixel signal corresponding to any one of the second or fourth pixel signal;

a third transistor that supplies the second electric signal to the pixel signal generation unit according to a third control signal; and

a fourth transistor that supplies the fourth electric signal to the pixel signal generation unit according to a fourth control signal.

REFERENCE SIGNS LIST

100 Imaging device
110 Imaging lens
120 Recording unit
130 Control unit
200 Solid-state imaging element
201 Light reception chip
202 Detection chip
211 Drive circuit
212 Signal processing unit
213 Arbiter
220 Column ADC
221 Reference signal generation unit
222 Output unit
230 ADC
231 Comparator
232 Counter
233 Switch
234 Memory
240 Differential amplification circuit
241, **242**, **412** P-type transistor
243, **244**, **245**, **411**, **413** N-type transistor
250 Counter
300 Pixel array unit
310 Pixel block
311 Pixel
312 Normal pixel
313 Address event detection pixel
320 Pixel signal generation unit
321 Reset transistor
322 Amplification transistor
323 Selection transistor
324 Floating diffusion layer
330 Light reception unit
331 Transfer transistor
332 OFG transistor
333 Photoelectric conversion element
400 Address event detection unit
410 Current/voltage conversion unit
420 Buffer
430 Subtractor
431, **433** Capacitor
432 Inverter
434 Switch
440 Quantizer
441 Comparator
450 Transfer unit
12031 Imaging unit

The invention claimed is:

1. A solid-state imaging device comprising:

a plurality of detection pixels that each output a luminance change of incident light;

a detection circuit that outputs an event signal based on the luminance change output from each of the detection pixels; and

a first common line connecting the plurality of detection pixels to each other,

wherein each of the detection pixels includes:

a photoelectric conversion element;

a logarithmic conversion circuit that converts a photocurrent flowing out of the photoelectric conversion element into a voltage signal corresponding to a logarithmic value of the photocurrent;

a first circuit that outputs a luminance change of incident light incident on the photoelectric conversion element based on the voltage signal output from the logarithmic conversion circuit;

a first transistor connected between the photoelectric conversion element and the logarithmic conversion circuit; and

a second transistor connected between the photoelectric conversion element and the first common line, wherein

the detection circuit includes a second circuit that outputs the event signal based on the luminance change output from each of the detection pixels.

2. The solid-state imaging device according to claim **1**, wherein the second transistor is connected between the first common line and a node, the node being configured to connect the photoelectric conversion element and the first transistor.

3. The solid-state imaging device according to claim **1**, further comprising:

a readout circuit that is connected to the first common line and that generates a pixel signal having a voltage value corresponding to charge accumulated in the photoelectric conversion element.

4. The solid-state imaging device according to claim **3**, wherein the readout circuit includes:

a reset transistor connected between the first common line and a power supply line; and

an amplification transistor having a gate connected to the first common line.

5. The solid-state imaging device according to claim **1**, wherein each of the detection pixels further includes a differentiator that generates a differential signal indicating a conversion amount of the voltage signal output from the logarithmic conversion circuit.

6. The solid-state imaging device according to claim **1**, wherein

the detection circuit is one of a plurality of detection circuits,

each of the detection circuits is configured to output a request for requesting readout of a detection signal from the detection circuit when having detected an address event in at least one of the plurality of detection pixels, and

an arbiter configured to arbitrate the request output from at least one of the plurality of detection circuits and determine a readout order of the detection signal for the detection circuit that has output the request.

7. The solid-state imaging device according to claim **1**, further comprising:

a first chip including a light receiving section, the light receiving section having a plurality of logarithmic response sections being arranged in a two-dimensional lattice pattern, each of the plurality of logarithmic response sections including the photoelectric conversion element, the logarithmic conversion circuit, the first transistor, and the second transistor.

8. The solid-state imaging device according to claim **7**, wherein the logarithmic conversion circuit includes:

a third transistor having a source connected to the first transistor; and a fourth transistor having a gate connected to the source of the third transistor and having a source grounded, the third transistor having a gate connected to a drain of the fourth transistor.

9. The solid-state imaging device according to claim **7**, wherein the light receiving section further includes a pixel isolation section extending in a lattice pattern, and

each of the logarithmic response sections is provided in each of pixel regions partitioned into the two-dimensional lattice pattern by the pixel isolation section.

10. The solid-state imaging device according to claim **9**, wherein the first and second transistors, at least two transistors different from the first and second transistors, and the photoelectric conversion element are disposed in the pixel region,

the at least two transistors are disposed at positions across the photoelectric conversion element in the pixel region, and

the logarithmic conversion circuit is constituted by using at least one of the at least two transistors in each of the two pixel regions adjacent to each other.

11. The solid-state imaging device according to claim **7**, wherein the detection circuit is one of a plurality of detection circuits, and further comprising a second chip on which a plurality of the detection circuits is disposed, wherein the first chip and the second chip constitute a single stacked chip.

12. An imaging device comprising:

the solid-state imaging device according to claim **1**; and

a control section that controls the solid-state imaging device.

13. The imaging device according to claim **12**, wherein the second transistor is connected between the first common line and a node, the node being configured to connect the photoelectric conversion element and the first transistor.

14. The imaging device according to claim **12**, further comprising:

a readout circuit that is connected to the first common line and that generates a pixel signal having a voltage value corresponding to charge accumulated in the photoelectric conversion element.

15. The imaging device according to claim **14**, wherein the readout circuit includes:

a reset transistor connected between the first common line and a power supply line; and

an amplification transistor having a gate connected to the first common line.

16. The imaging device according to claim **12**, wherein each of the detection pixels further includes a differentiator that generates a differential signal indicating a conversion amount of the voltage signal output from the logarithmic conversion circuit.

17. The imaging device according to claim **12**, wherein

the detection circuit is one of a plurality of detection circuits,

each of the detection circuits is configured to output a request for requesting readout of a detection signal from the detection circuit when having detected an address event in at least one of the plurality of detection pixels, and

an arbiter configured to arbitrate the request output from at least one of the plurality of detection circuits and determine a readout order of the detection signal for the detection circuit that has output the request.

18. The imaging device according to claim **12**, further comprising:

a first chip including a light receiving section, the light receiving section having a plurality of logarithmic response sections being arranged in a two-dimensional lattice pattern, each of the plurality of logarithmic response sections including the photoelectric conversion element, the logarithmic conversion circuit, the first transistor, and the second transistor.

19. The imaging device according to claim **18**, wherein the logarithmic conversion circuit includes:

a third transistor having a source connected to the first transistor; and

a fourth transistor having a gate connected to the source of the third transistor and having a source grounded, the third transistor having a gate connected to a drain of the fourth transistor.

20. The imaging device according to claim **18**,

wherein the light receiving section further includes a pixel isolation section extending in a lattice pattern, and

each of the logarithmic response sections is provided in each of pixel regions partitioned into the two-dimensional lattice pattern by the pixel isolation section.

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